



PDPM IIITDM JABALPUR
MECHANICAL ENGINEERING DISCIPLINE

Computational Fluid Dynamics Analysis of Heat Sink with Phase Change Material

Presented by:

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Overview

01 Introduction

02 Problem Statement

03 Literature Review

04 Methodology

05 Result & Discussion

06 Validation & Conclusion

Introduction



- ❑ The rising need for high-performance electronic gadgets in recent years has resulted in compact systems producing more heat and having higher power densities.
- ❑ Effective heat control is now essential to for these equipment to increase the dependability and durability.
- ❑ Heat sinks are one of the most promising solutions in thermal management.
- ❑ In order to overcome this difficulty, this study examines the functionality of a heat sink that incorporates Phase Change Materials (PCMs) for improved thermal control.

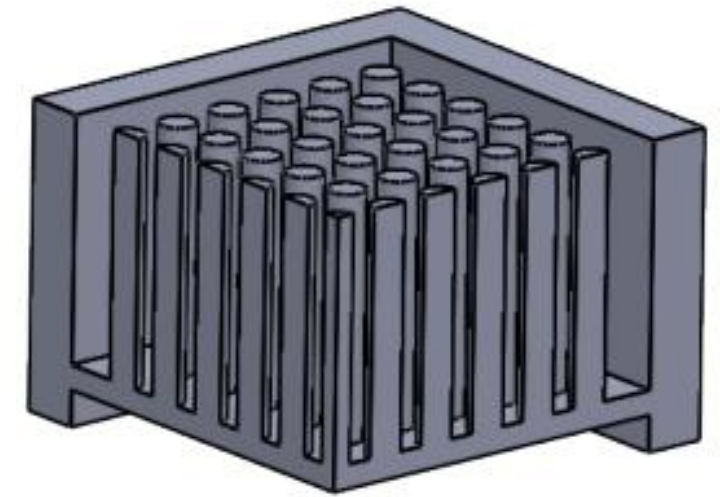


Fig. heat sink



Problem Statement

- ❑ Advanced electronics generate excessive heat, requiring efficient thermal management to prevent overheating and ensure reliability.
- ❑ Traditional heat sinks are inadequate for modern high-performance devices, prompting the exploration of Phase Change Materials (PCMs) for enhanced cooling.
- ❑ The study focuses on including PCM at higher power levels to improve heat sink efficiency.

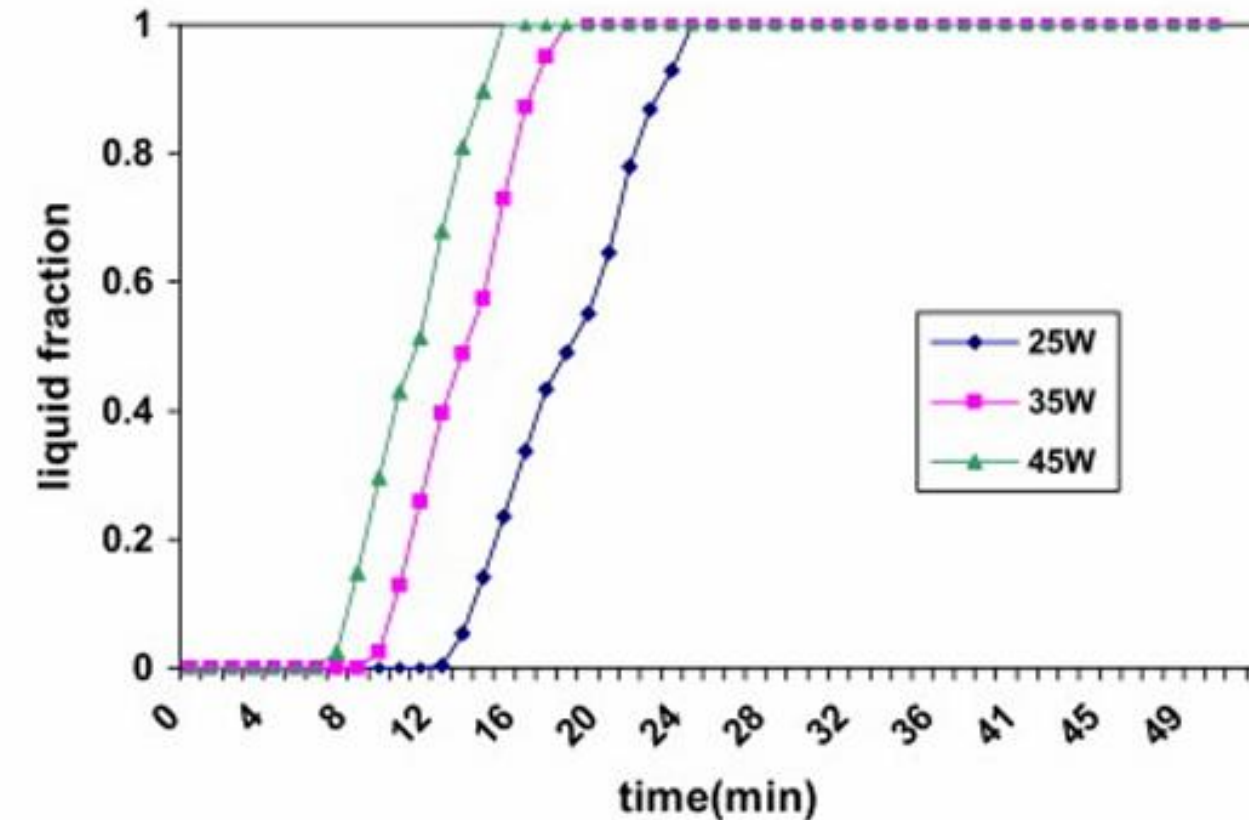
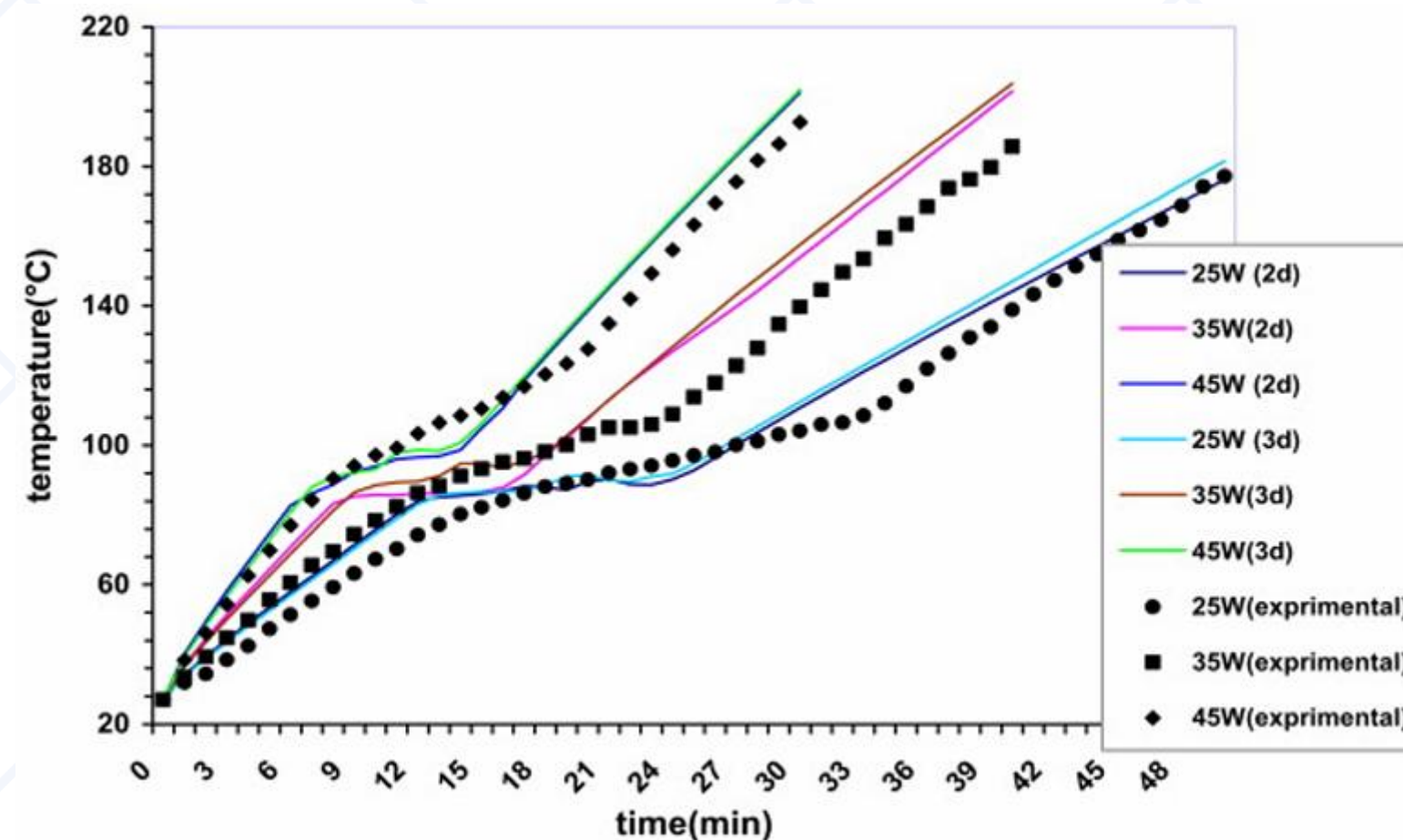
Objective

- ❑ Design and Model Development
- ❑ Thermal Performance Analysis
- ❑ PCM Behavior Evaluation
- ❑ Contour Plot Visualization
- ❑ Comparison and Validation

Literature Review

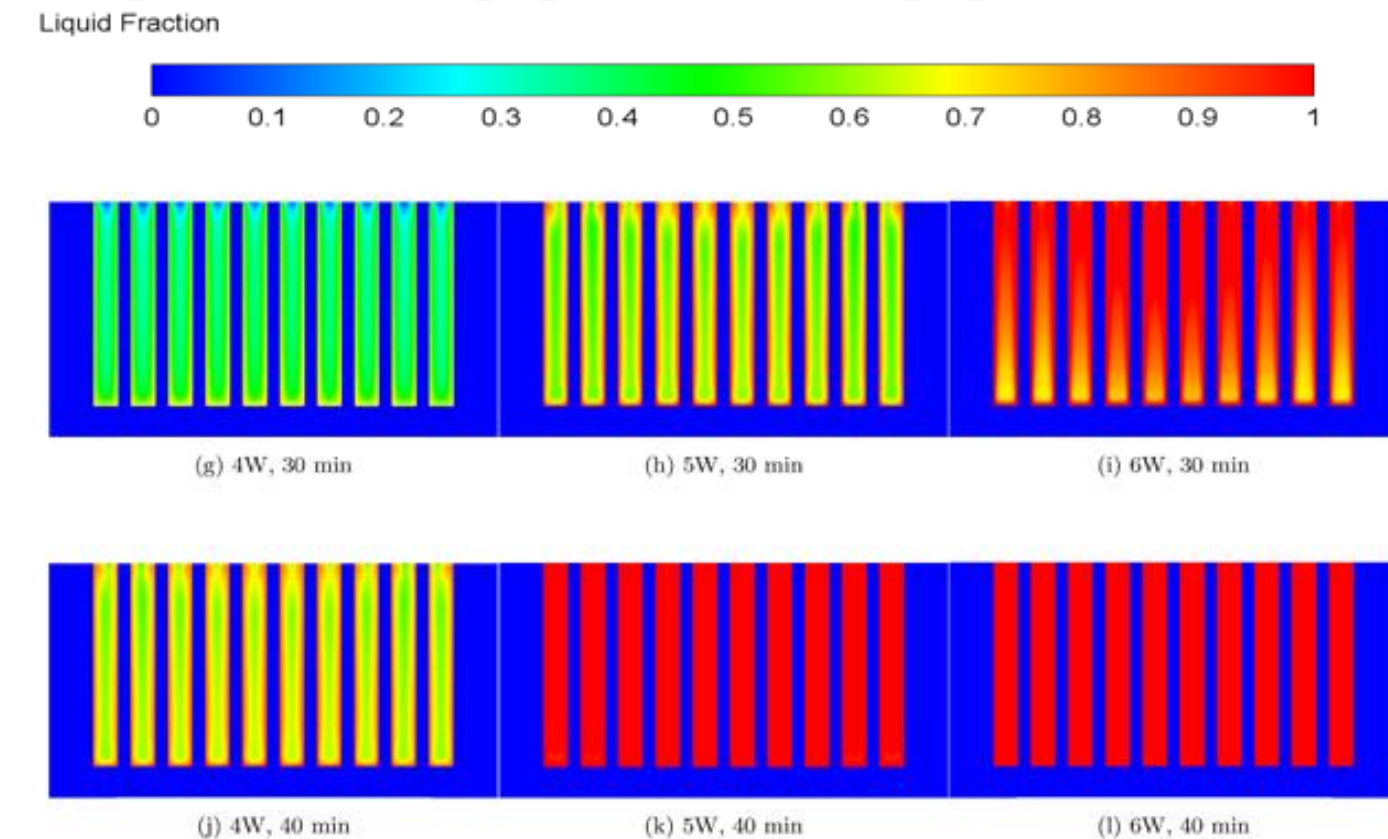
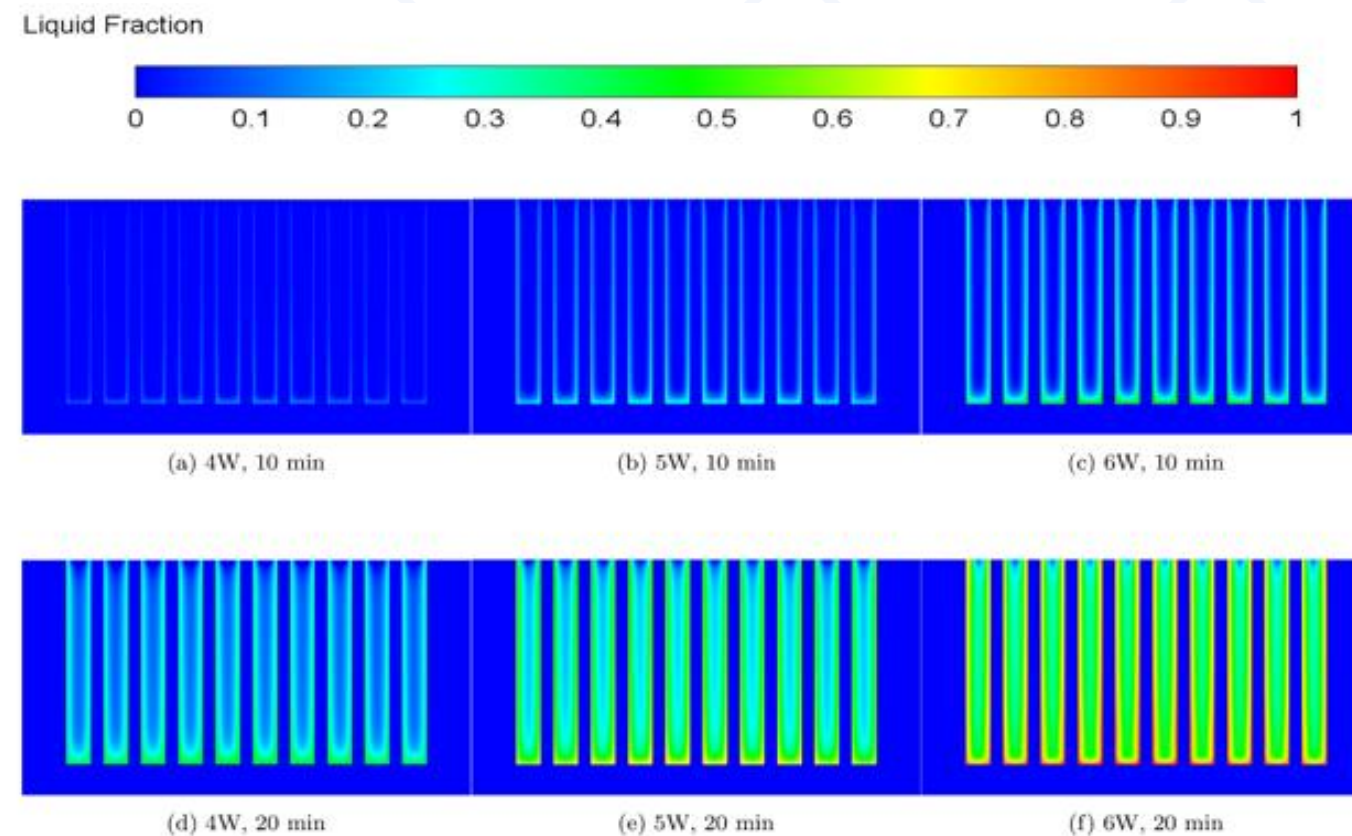


- [1] In this paper Shatikian et al. numerically studied a PCM-based heat sink with internal fins to enhance thermal management in electronics. The PCM absorbs latent heat, while fins improve heat transfer. Simulations analyzed temperature distribution, heat transfer rates, and performance. The findings highlight the effectiveness of PCM-fin heat sinks, providing a foundation for future research.

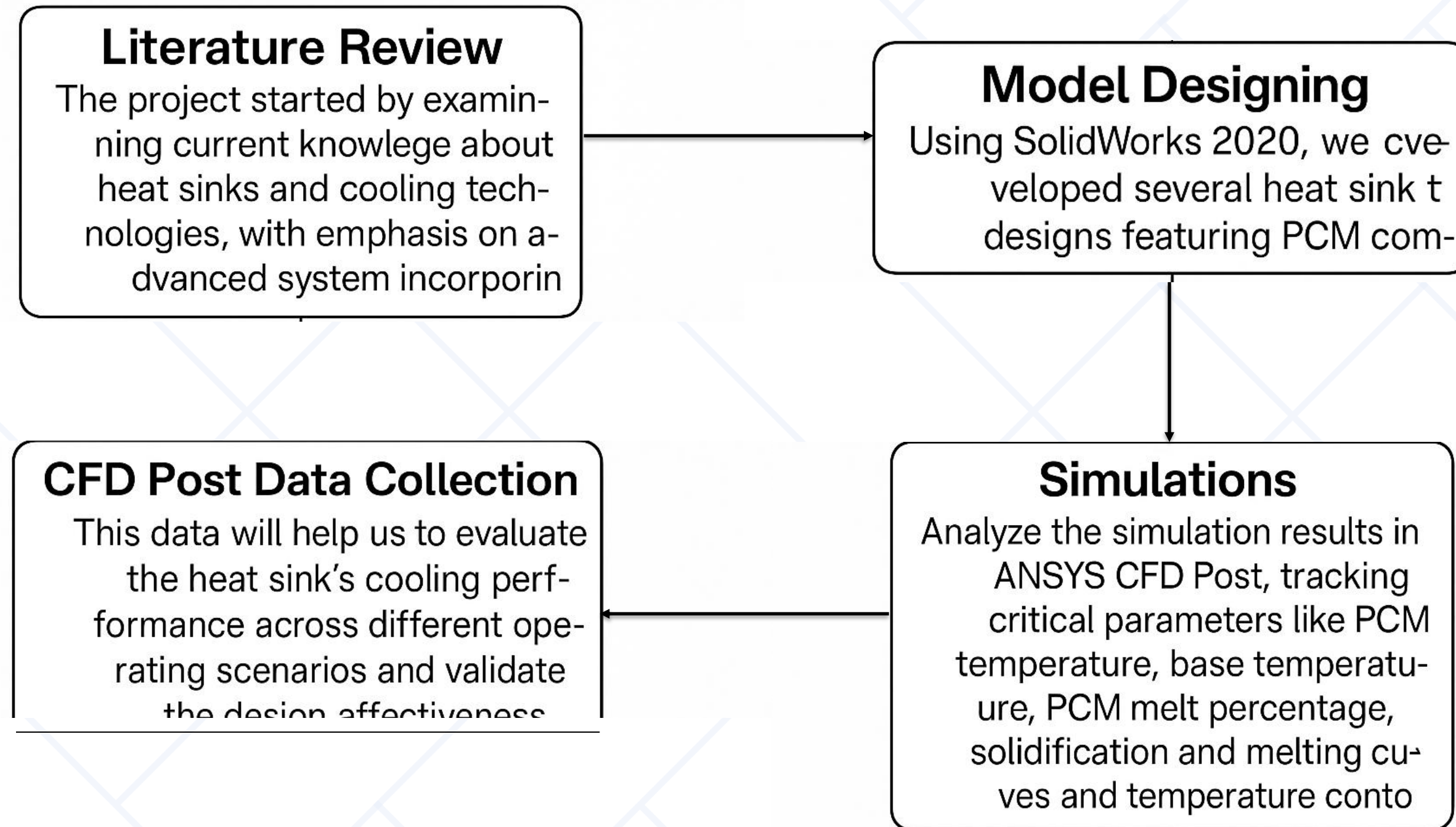




- ❑ [2] Arshad et al. studied PCM-based finned heat sinks (n-eicosane, Al fins) for electronics cooling using ANSYS Fluent. Thicker (3 mm) fins maintained lower base temps despite slower melting. Higher power accelerated PCM melting. Dimensionless correlations linked liquid fraction and Nusselt number to key parameters, demonstrating PCM-fin synergy for improved thermal management.



Methodology



Key Parameters

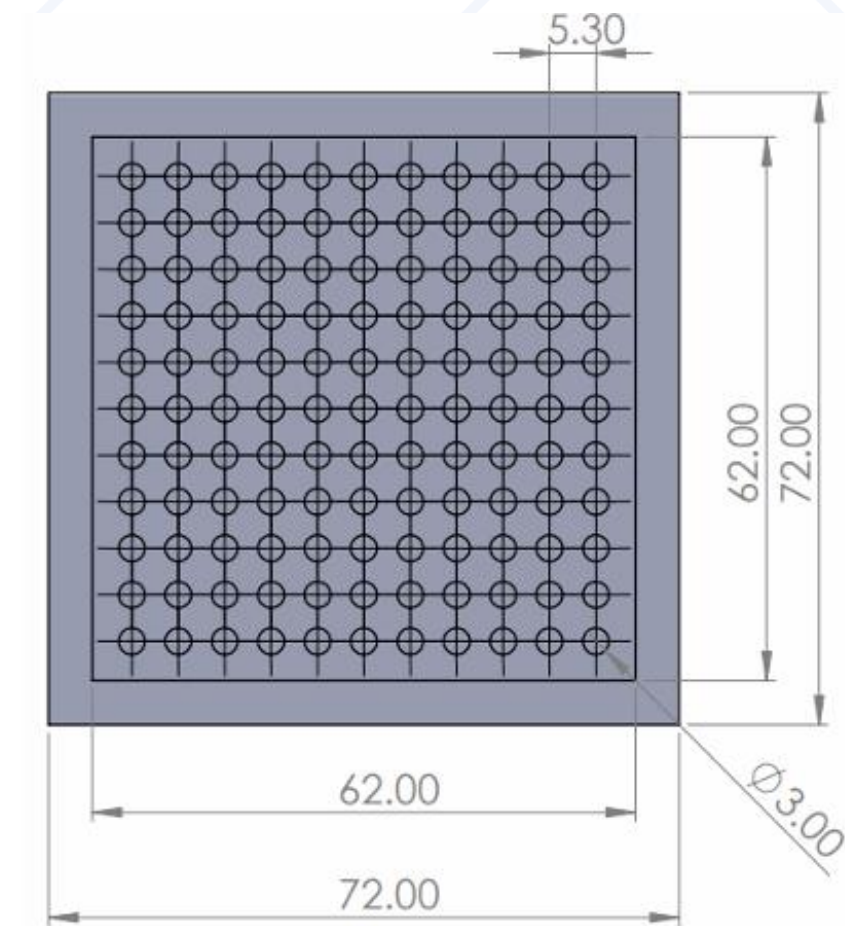


Dimensions of Heat Sink are mentioned in given below table :

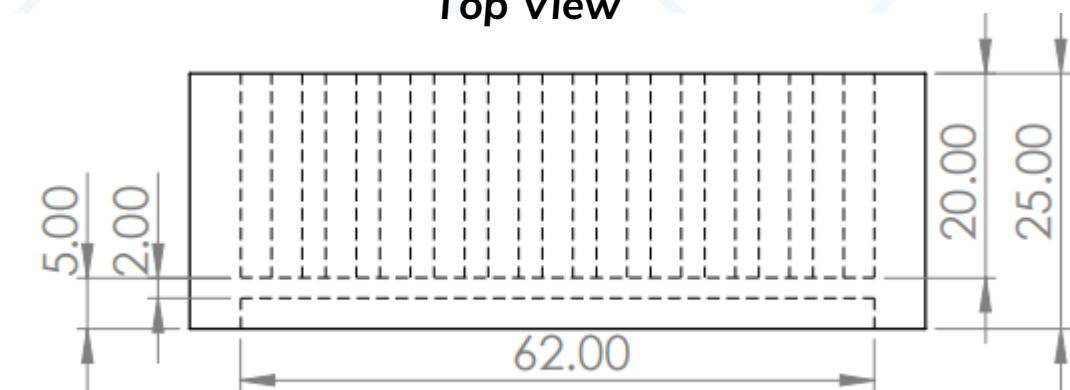
Components	(mm)
Heat Sink	75 x 75 x 25
Fin's Diameter	3
Fin's Height	20
Fin's Gap	5.3

Material properties used for the testing are mentioned below :

Material	Density(kg/m ³)	Specific Heat(J/kg K)	Thermal Conductivity (W/mK)
Aluminum	2719	871	202.4
Air	1.225	1006.43	0.0242
PCM	790	2050	0.23



Top View



Front View

Fig. Cad model of heat sink

Key Parameters



Properties of n eicosane ($C_{20}H_{42}$) phase change material:

PCM properties	
Density(boussinesq)	790 kg/m ³
Viscosity	0.00355 kg/m-s
Solidus Temperature	308.5 K
Specific heat	2050 J/ kgK
Thermal conductivity	0.23 W/mK
Melting Heat	241000 J/kg
Liquidus Temperature	310.5 K

1. High Latent heat
2. High Melting point
3. Suitable for low-temperature thermal storage
4. Chemically inert, non-toxic, and non-corrosive

Advantages:

- ✓ High energy storage density (stores large amounts of heat per unit mass)
- ✓ Good thermal stability (can undergo thousands of cycles without degradation)
- ✓ Low supercooling tendency (freezes near its melting point)
- ✓ Compatibility with containers (does not react with most metals/polymers)



Boundary Conditions :

Boundary	Conditions
Air-top Convection	20 W/m ² K
Fin-Top Convection	20 W/m ² K
Heat Input	25W, 30W and 35W
Side-wall convection	18 W/m ²
Wall-Top Convection	20 W/m ²

Number of Nodes	Number of Elements
120740	95175

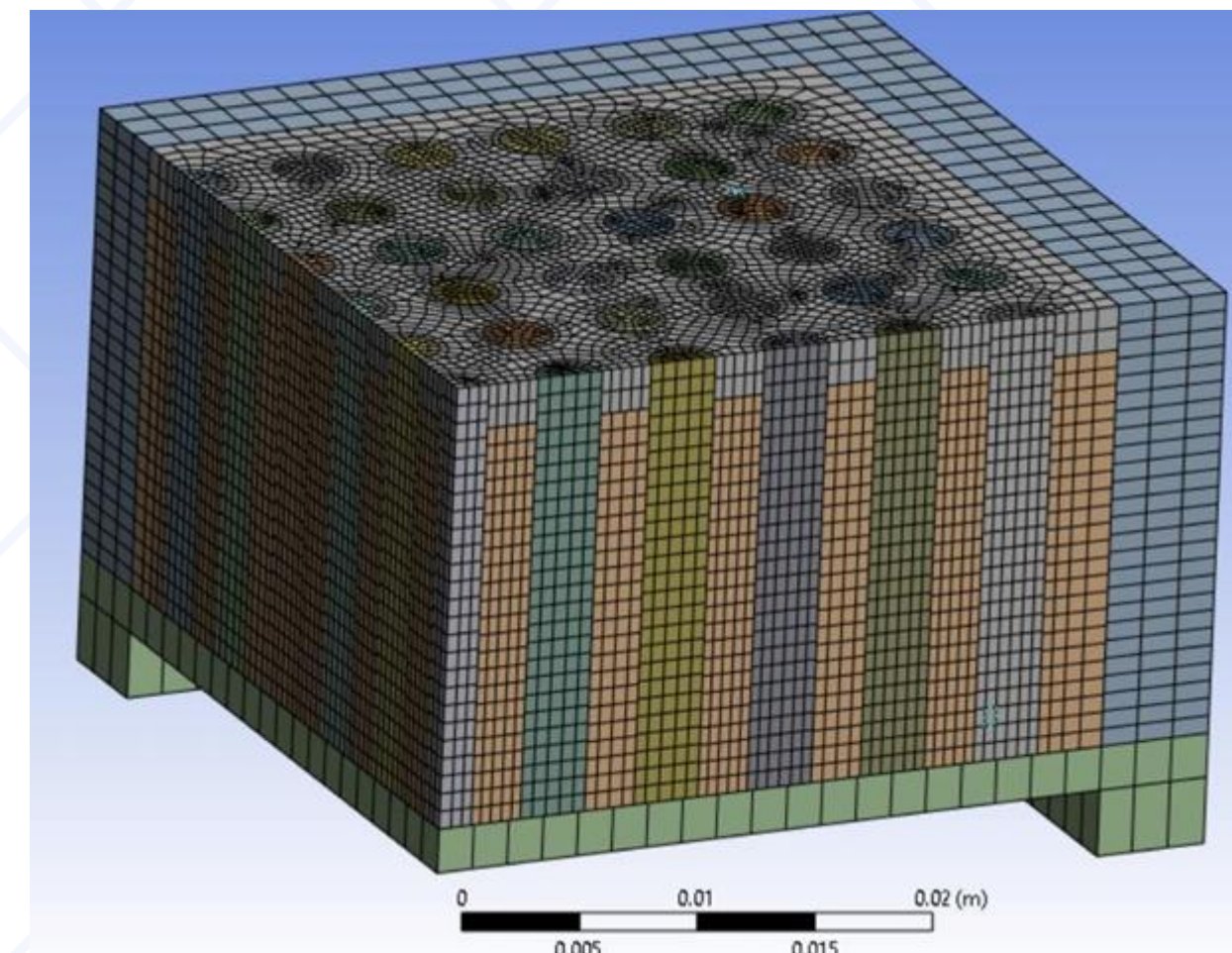


Fig. hex dominant meshing



Meshing Methods:

The various meshing pattern applied on the model are :

- ☐ Sweep Method on Fins with 25 numbers of divisions.
- ☐ Sweep Method on Side Walls with 25 divisions.
- ☐ Sweep Method on PCM with 23 divisions.
- ☐ Edge sizing on various edges with different parameters such as 19,22 divisions.
- ☐ Face Meshing on the base plate of the heat sink.
- ☐ Multizone Meshing on the base aluminum body of the heat sink.

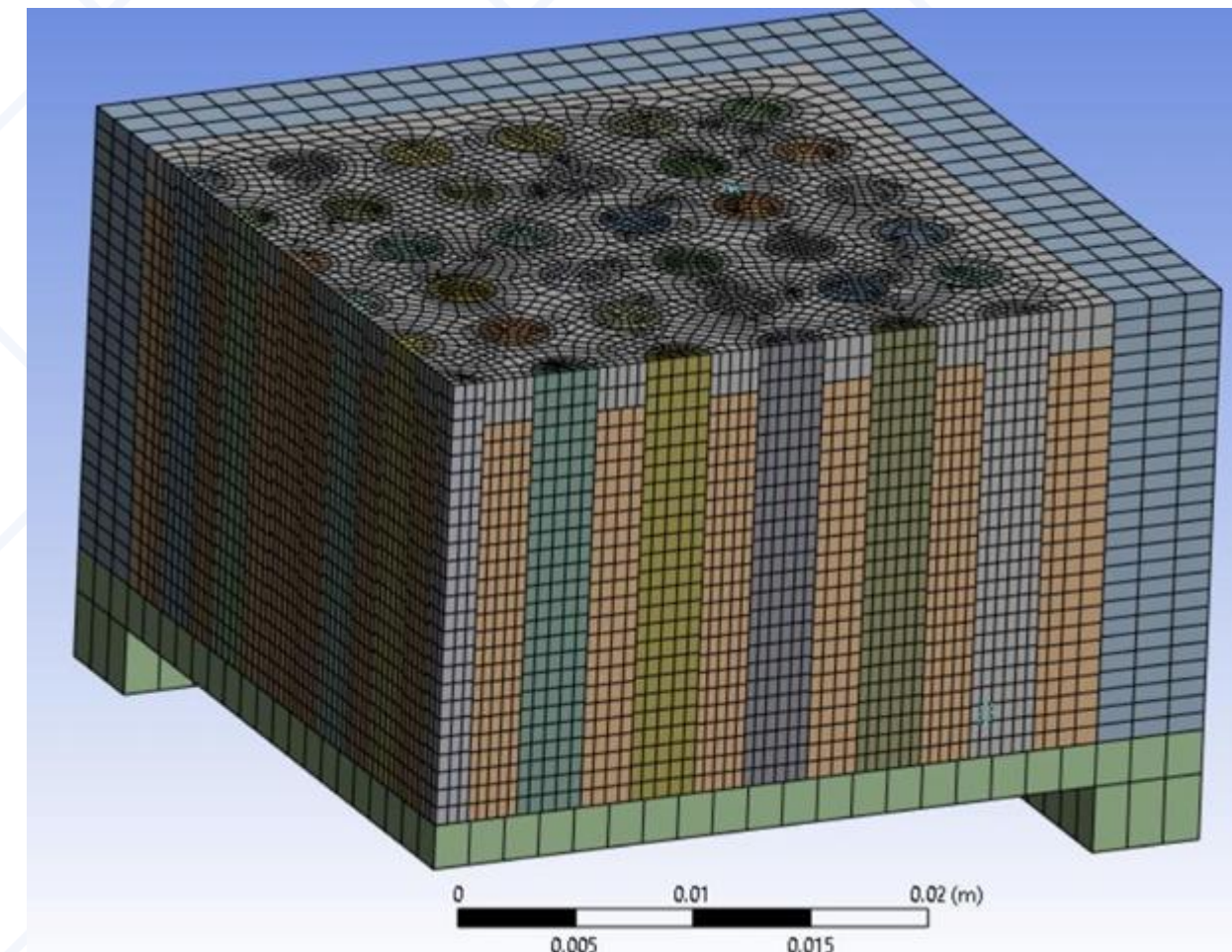


Fig. hex dominant meshing



Result & Discussion

(1) THE CYLINDRICAL PIN FIN CONFIGURATION AT 25W:

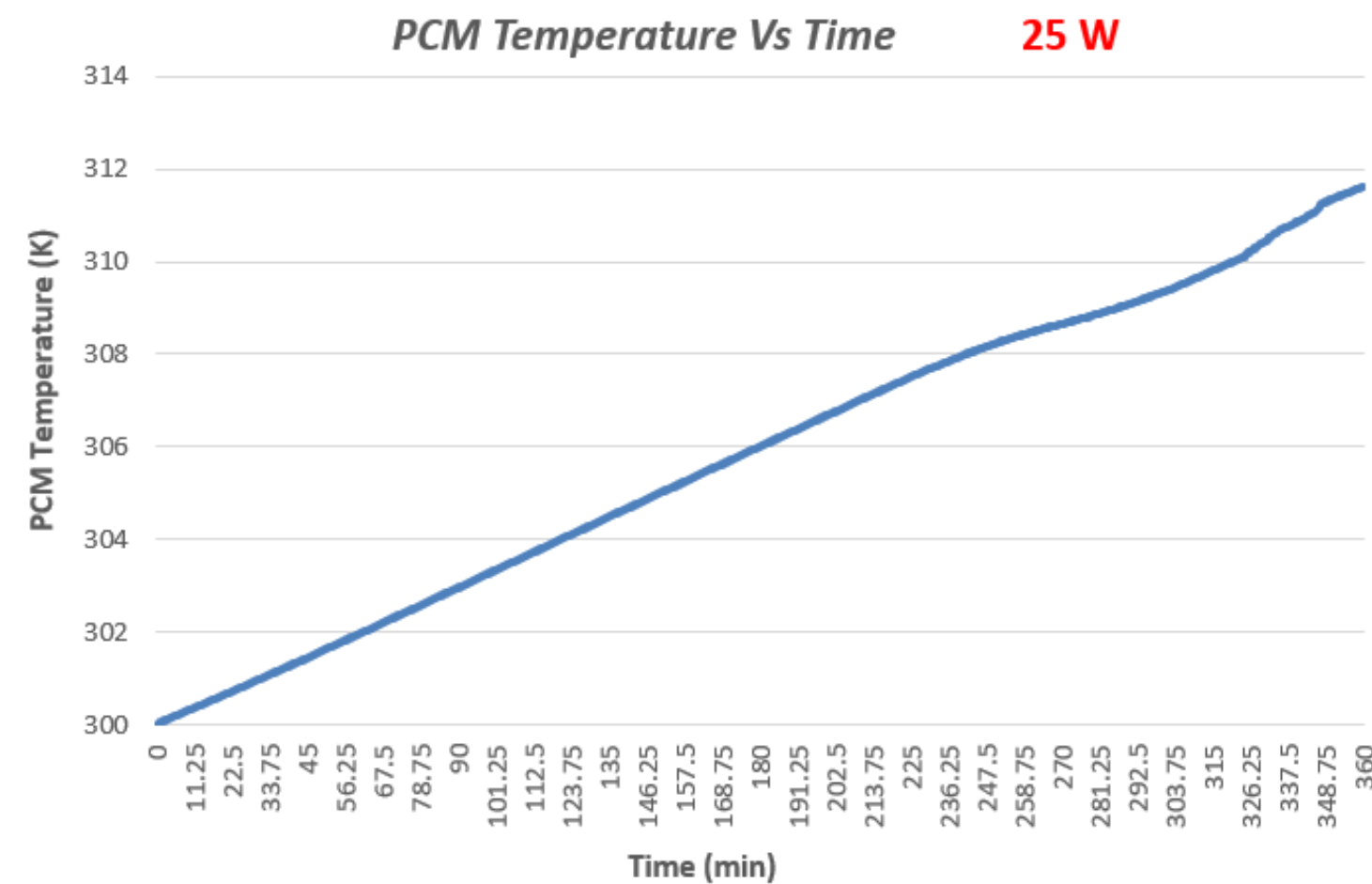


Fig. PCM Temperature vs Time graph

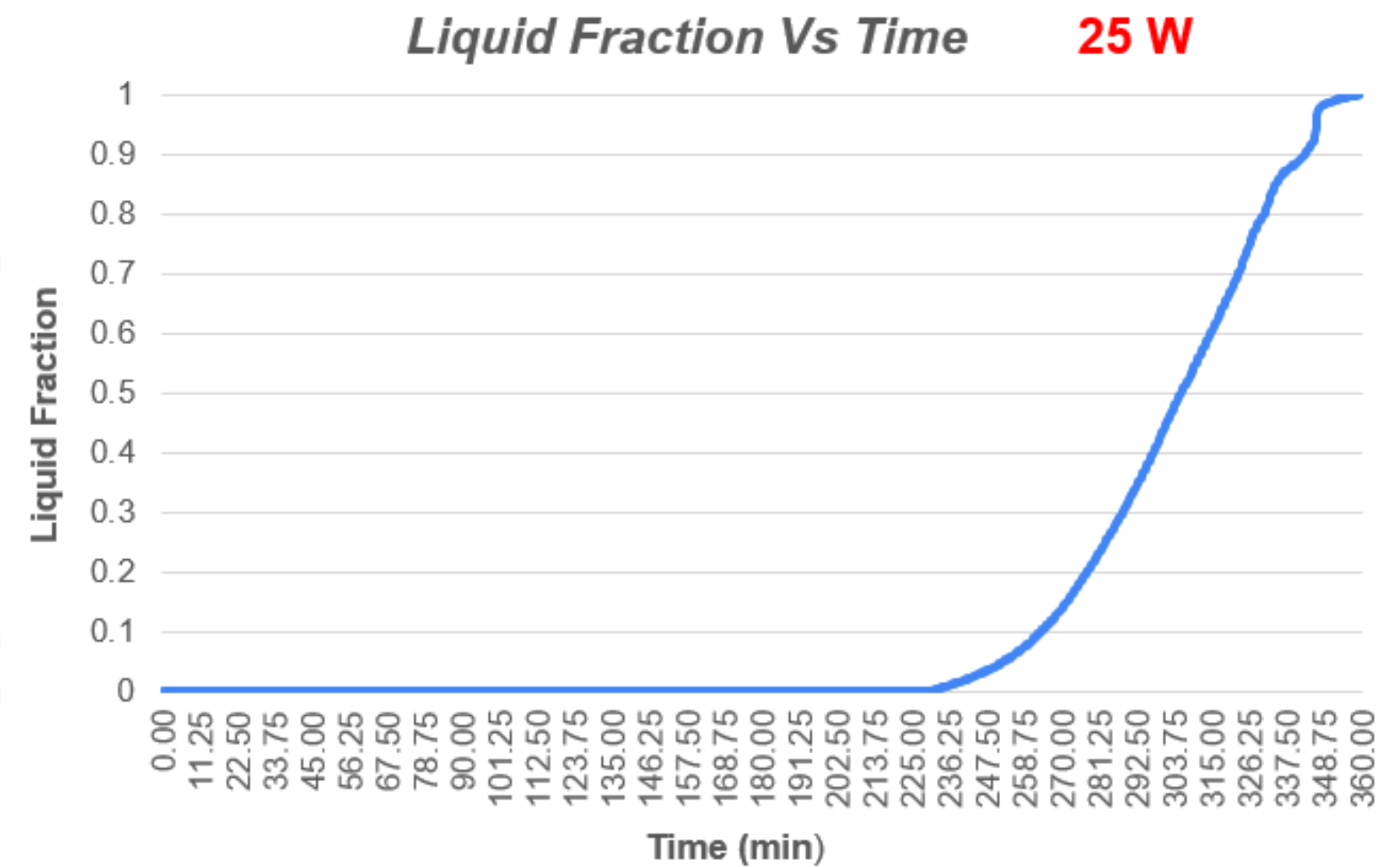


Fig. Liquid Fraction vs Time graph



Liquid Fraction Contours:

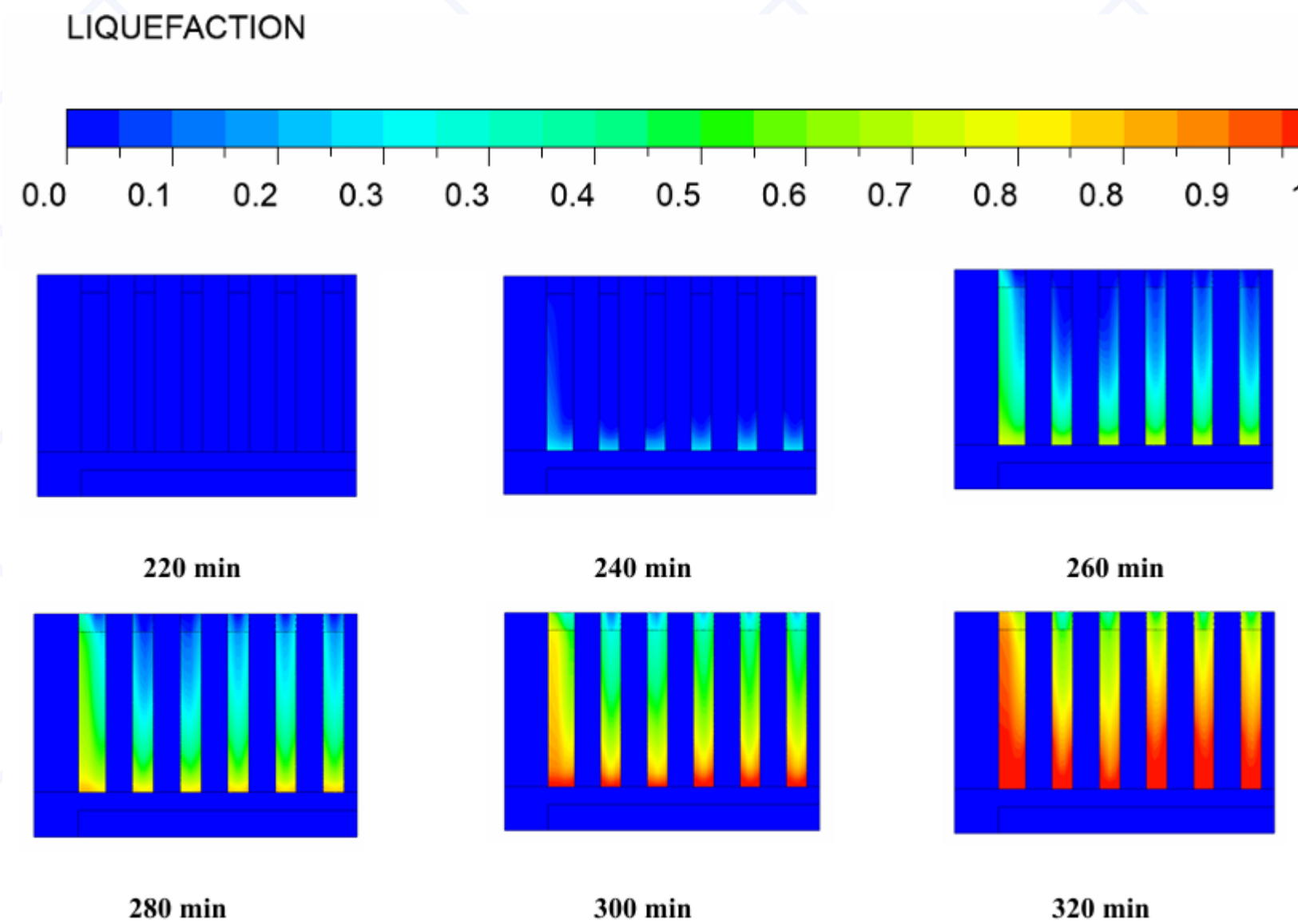


Fig. Liquid Fraction Contours (2D)

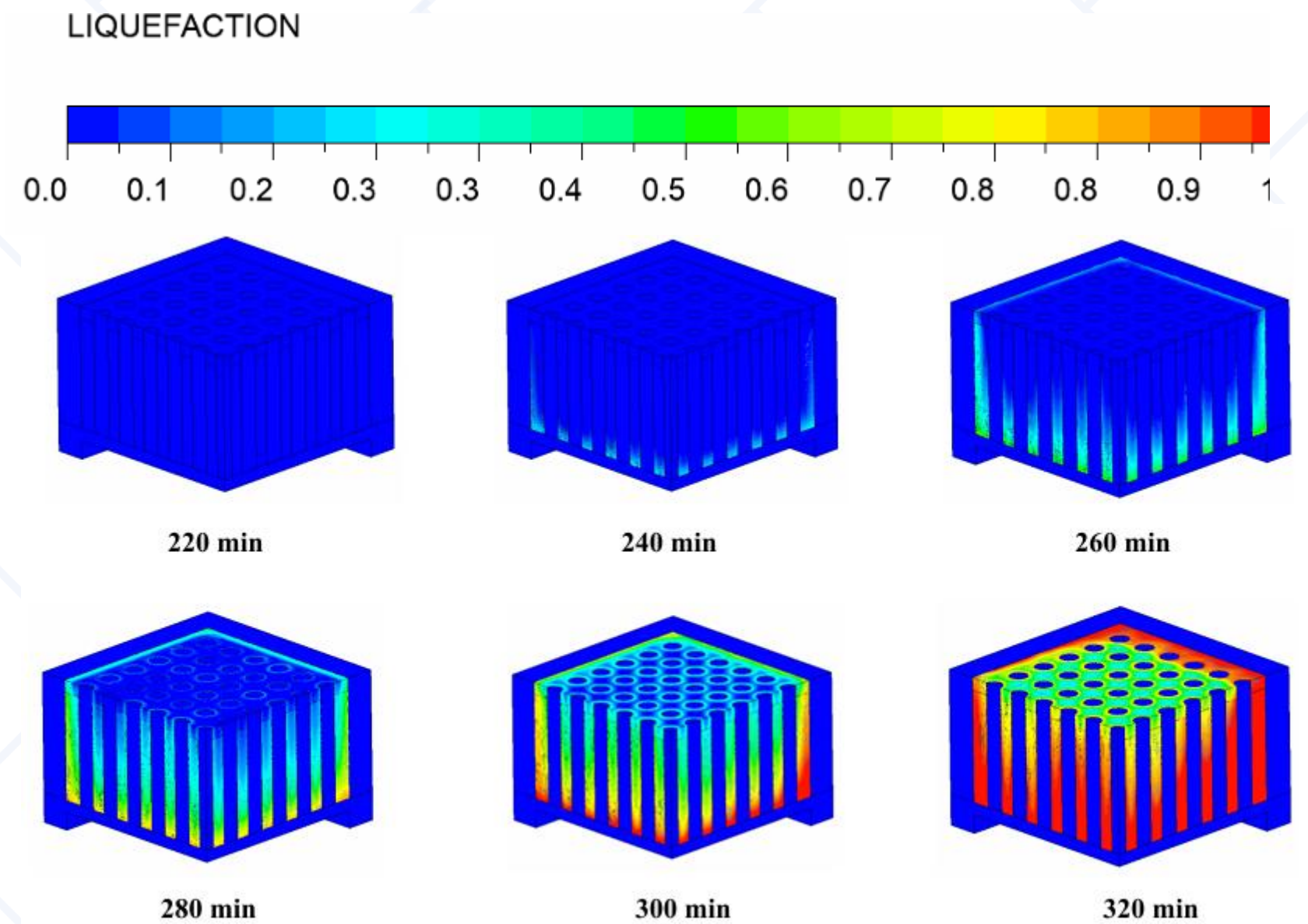


Fig. Liquid Fraction Contours (3D)



Temperature Contours:

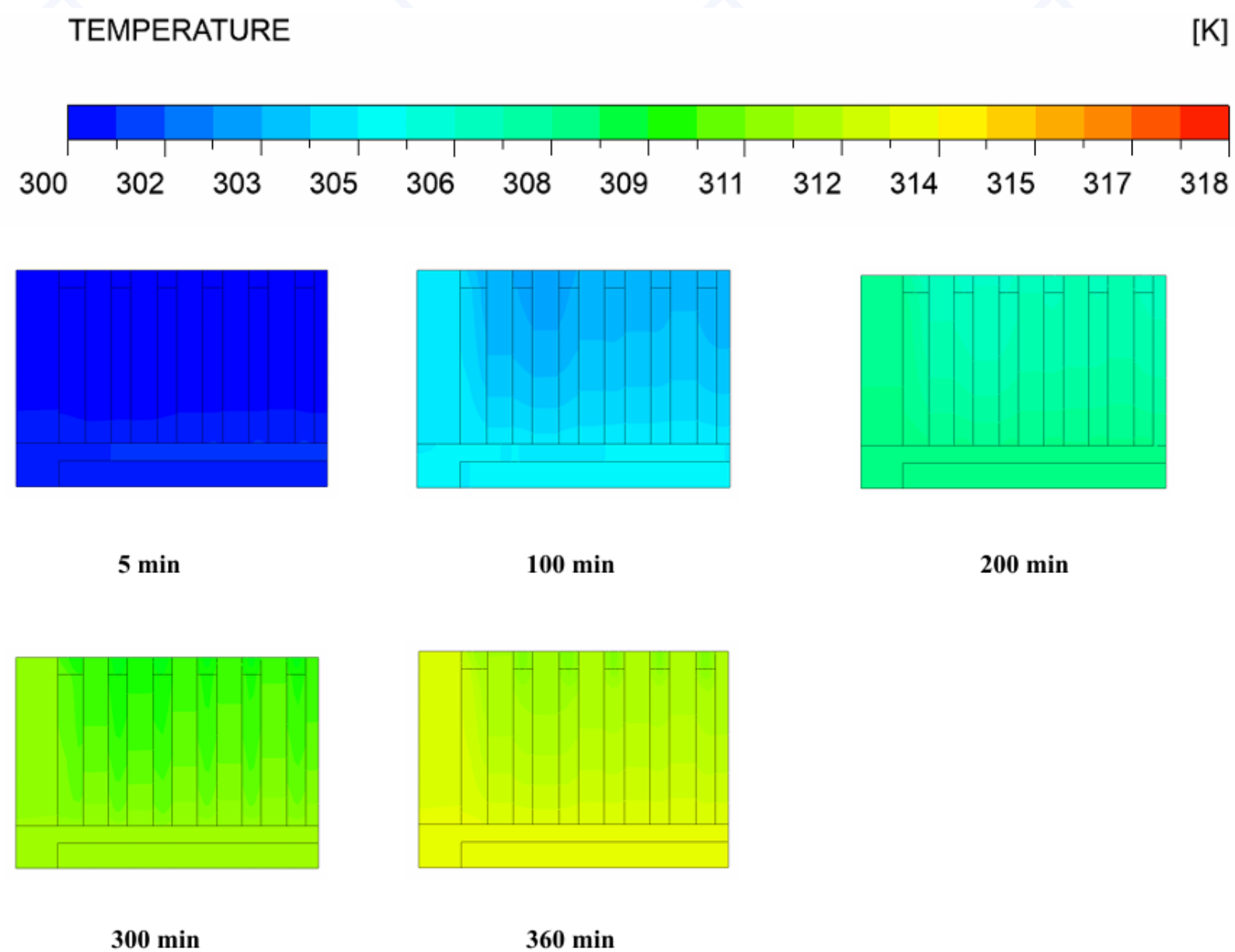


Fig. Temperature Contours (2D)

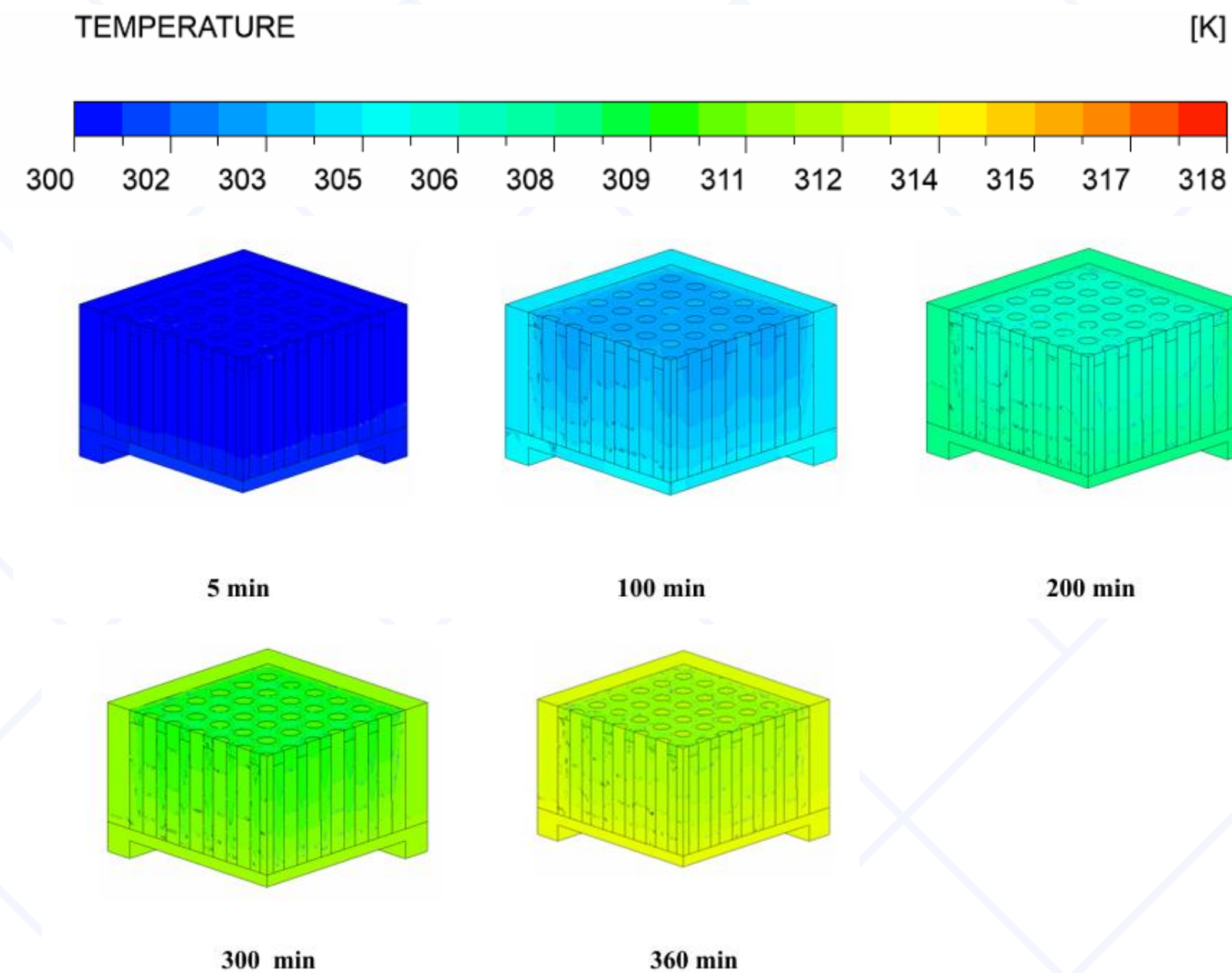


Fig. Temperature Contours (3D)



(2) THE CYLINDRICAL PIN FIN CONFIGURATION AT 30W:

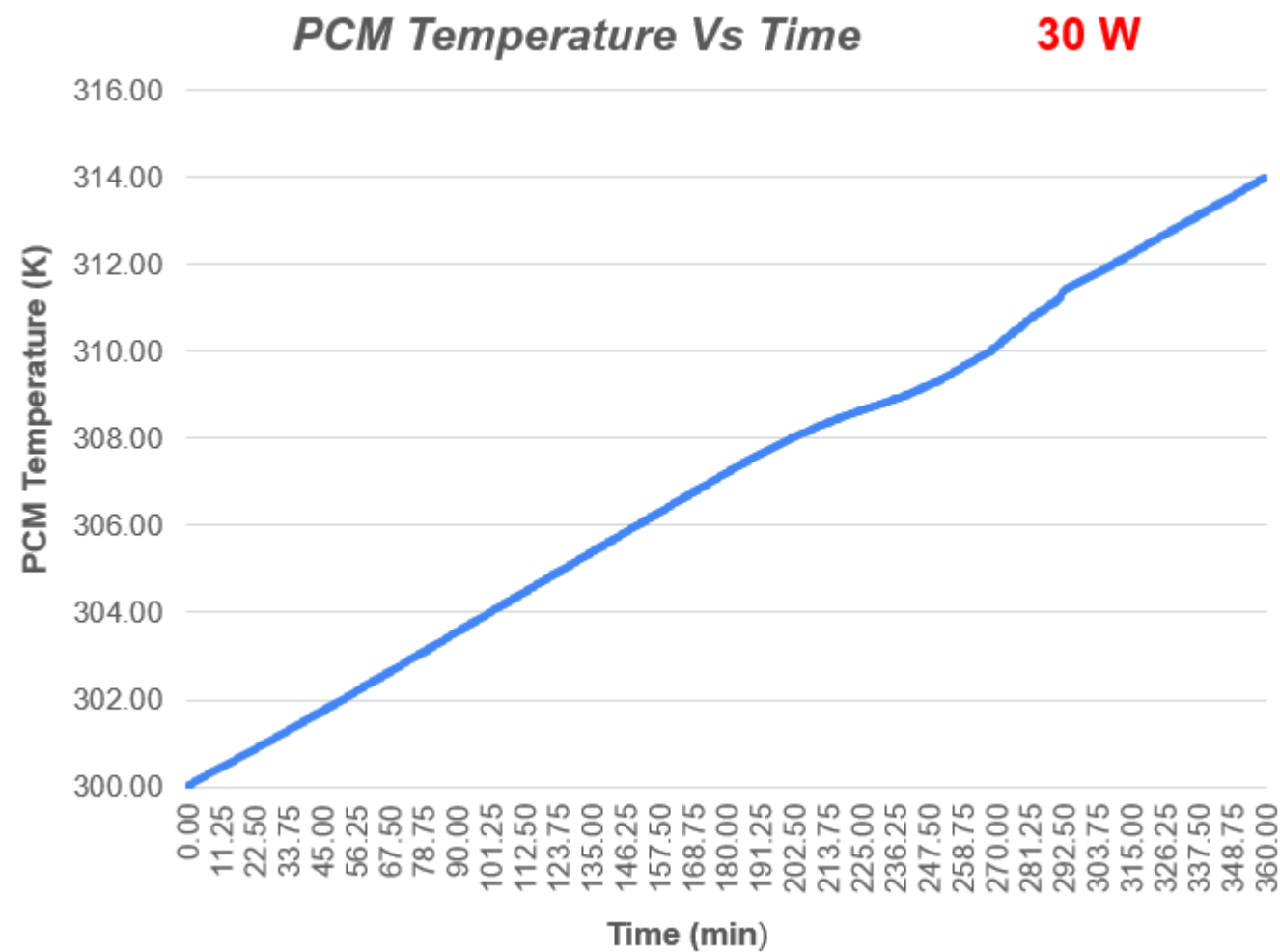


Fig. PCM Temperature vs Time graph

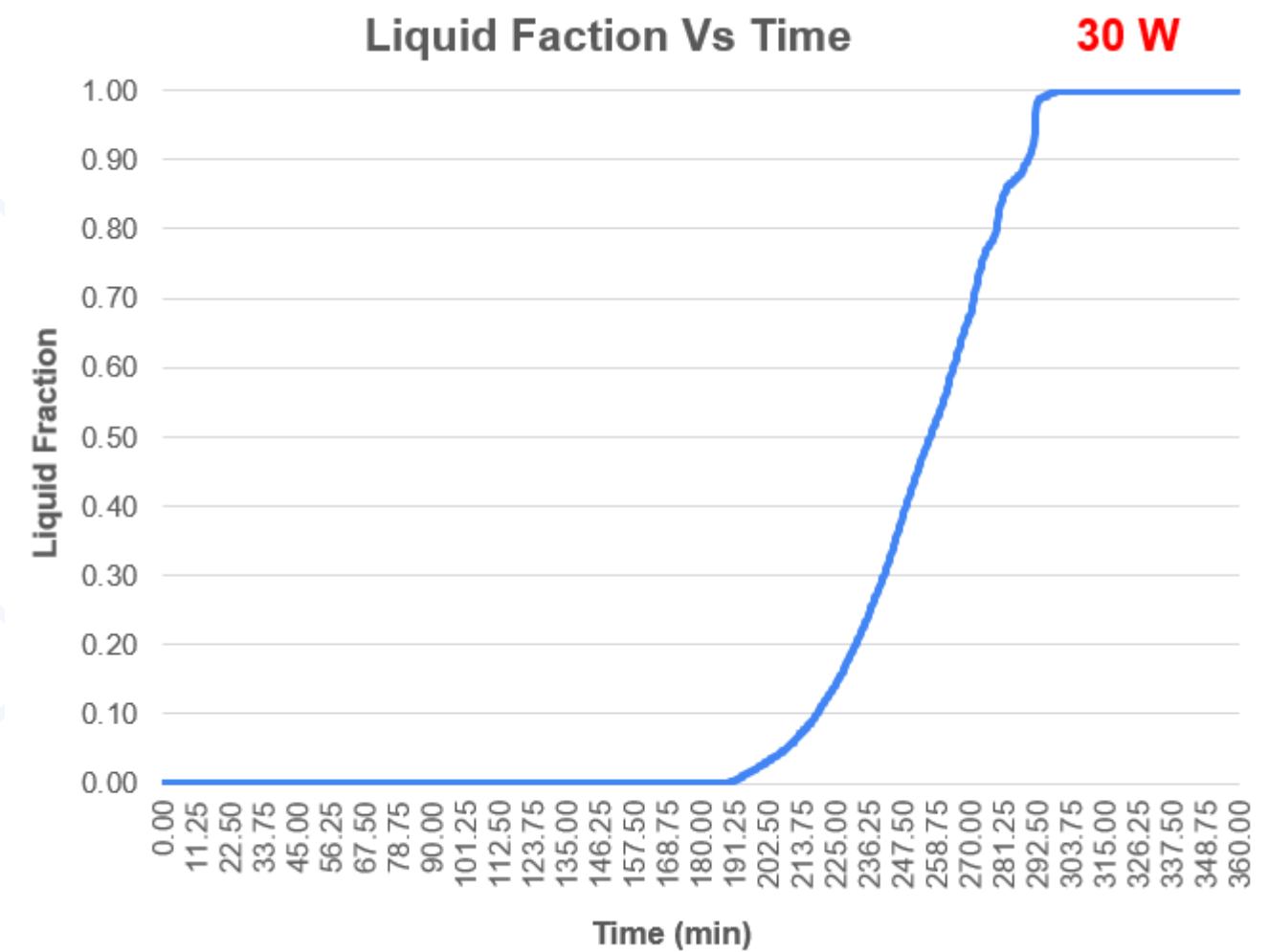


Fig. Liquid Fraction vs Time graph



Liquid Fraction Contours:

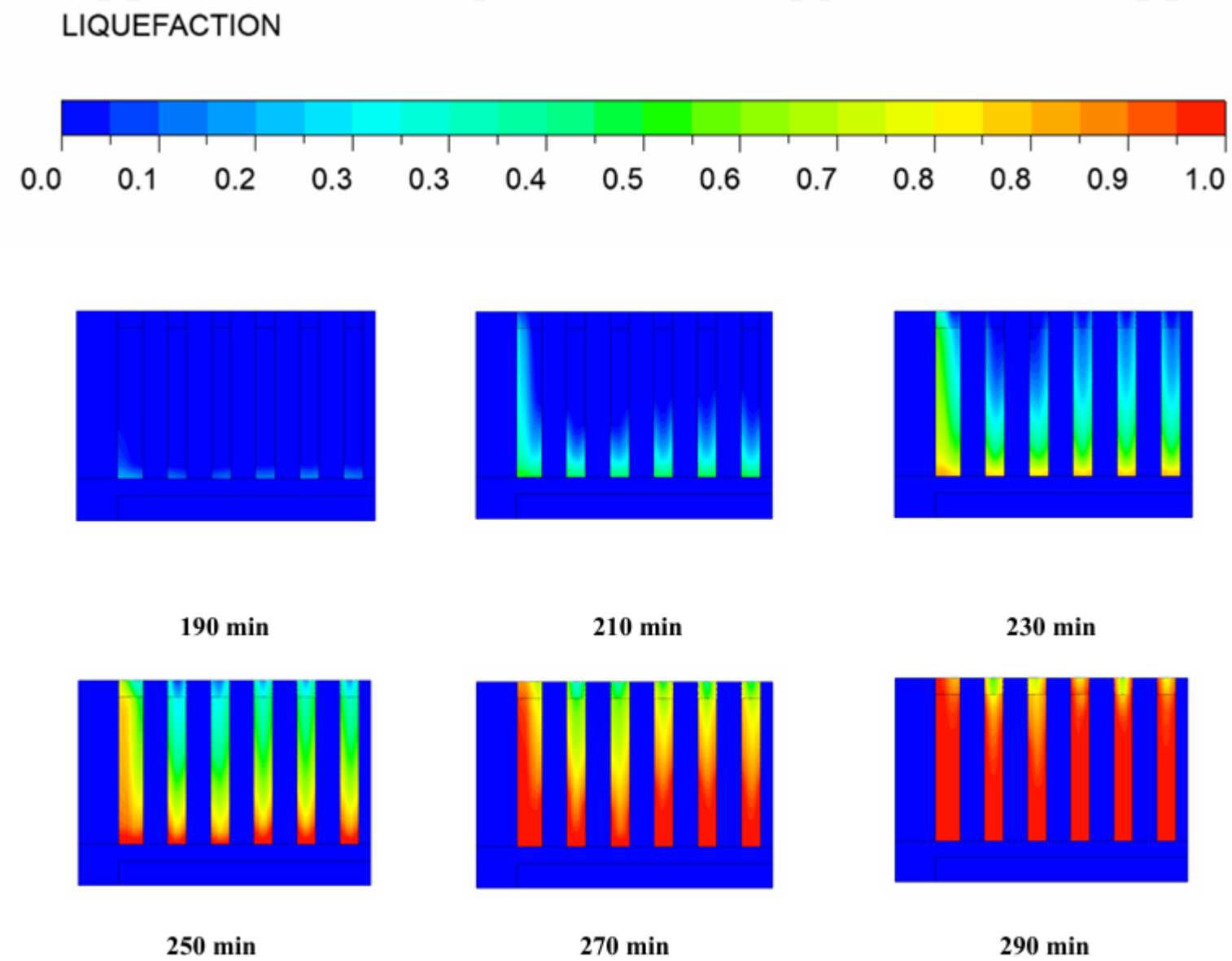


Fig. Liquid Fraction Contours (2D)

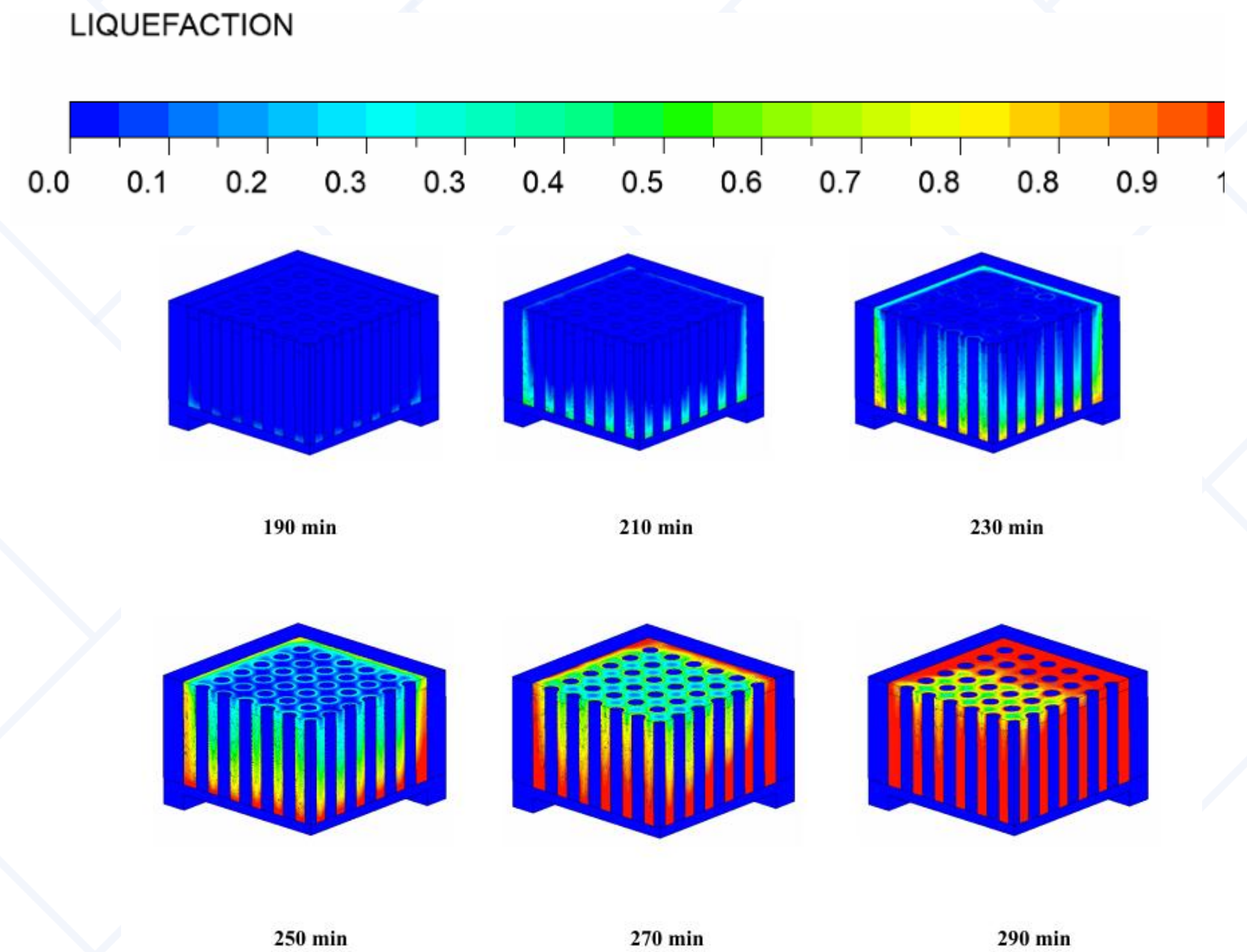


Fig. Liquid Fraction Contours (3D)



Temperature Contours:

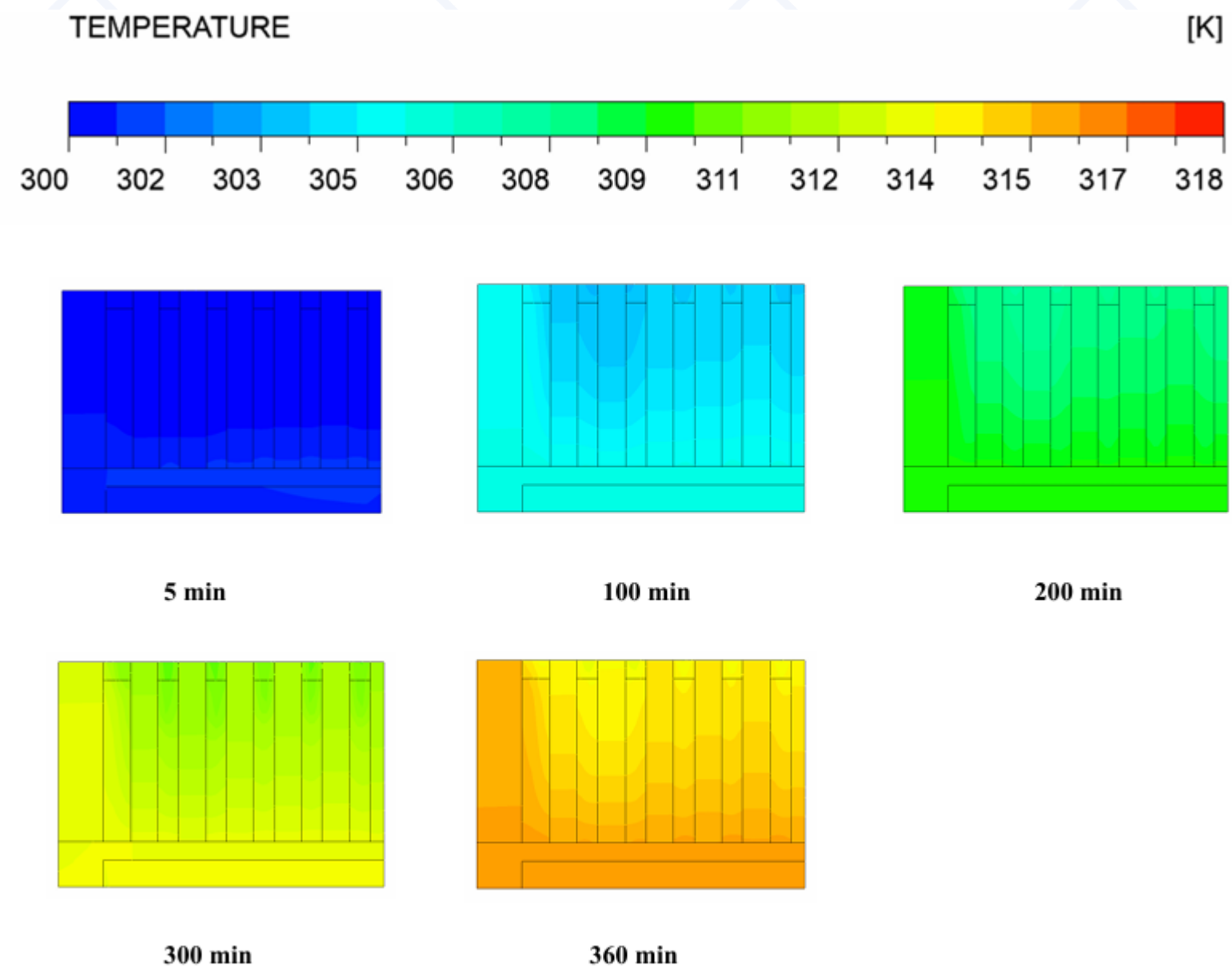


Fig. Temperature Contours (2D)

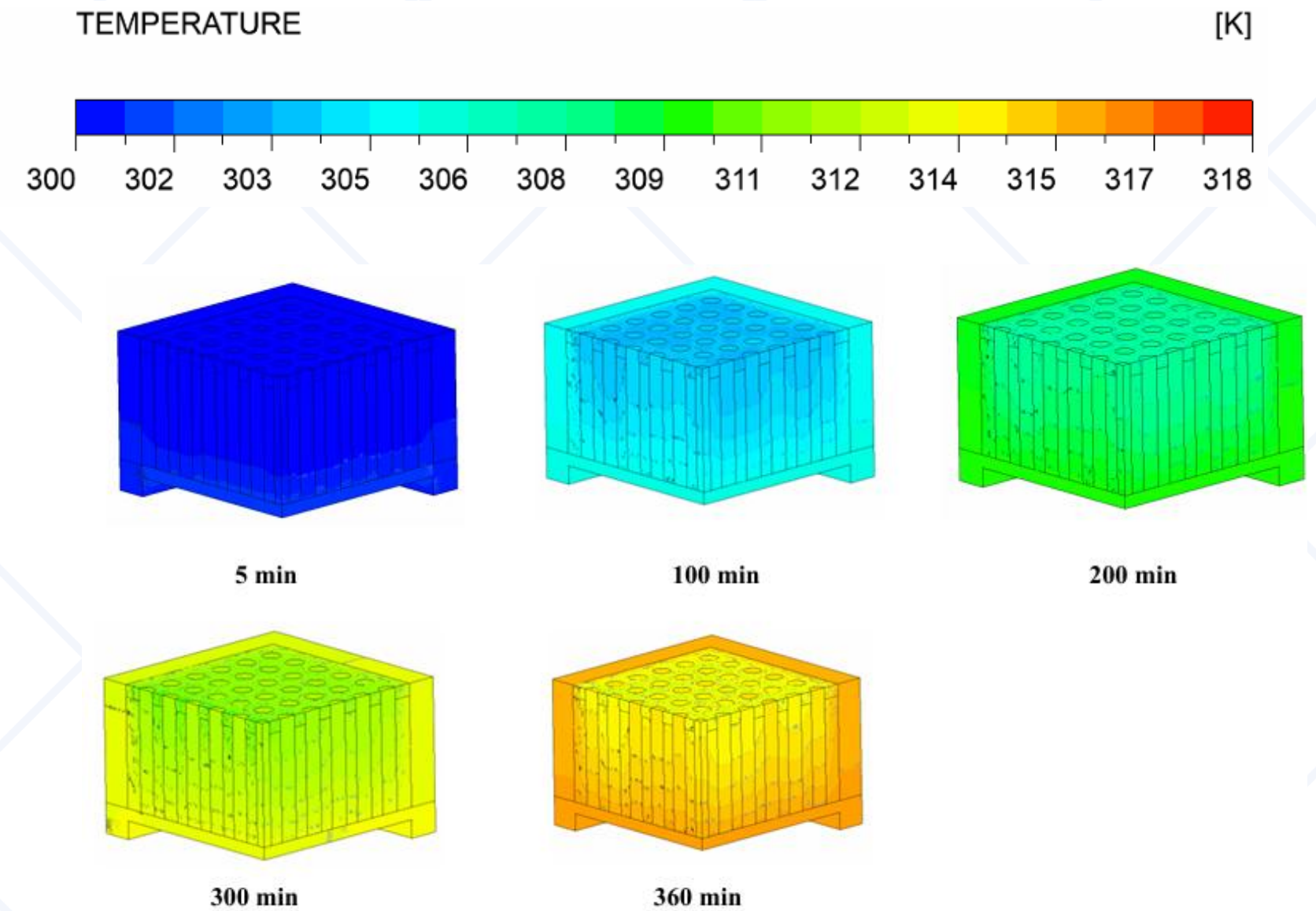


Fig. Temperature Contours (3D)



(3) THE CYLINDRICAL PIN FIN CONFIGURATION AT 35W:

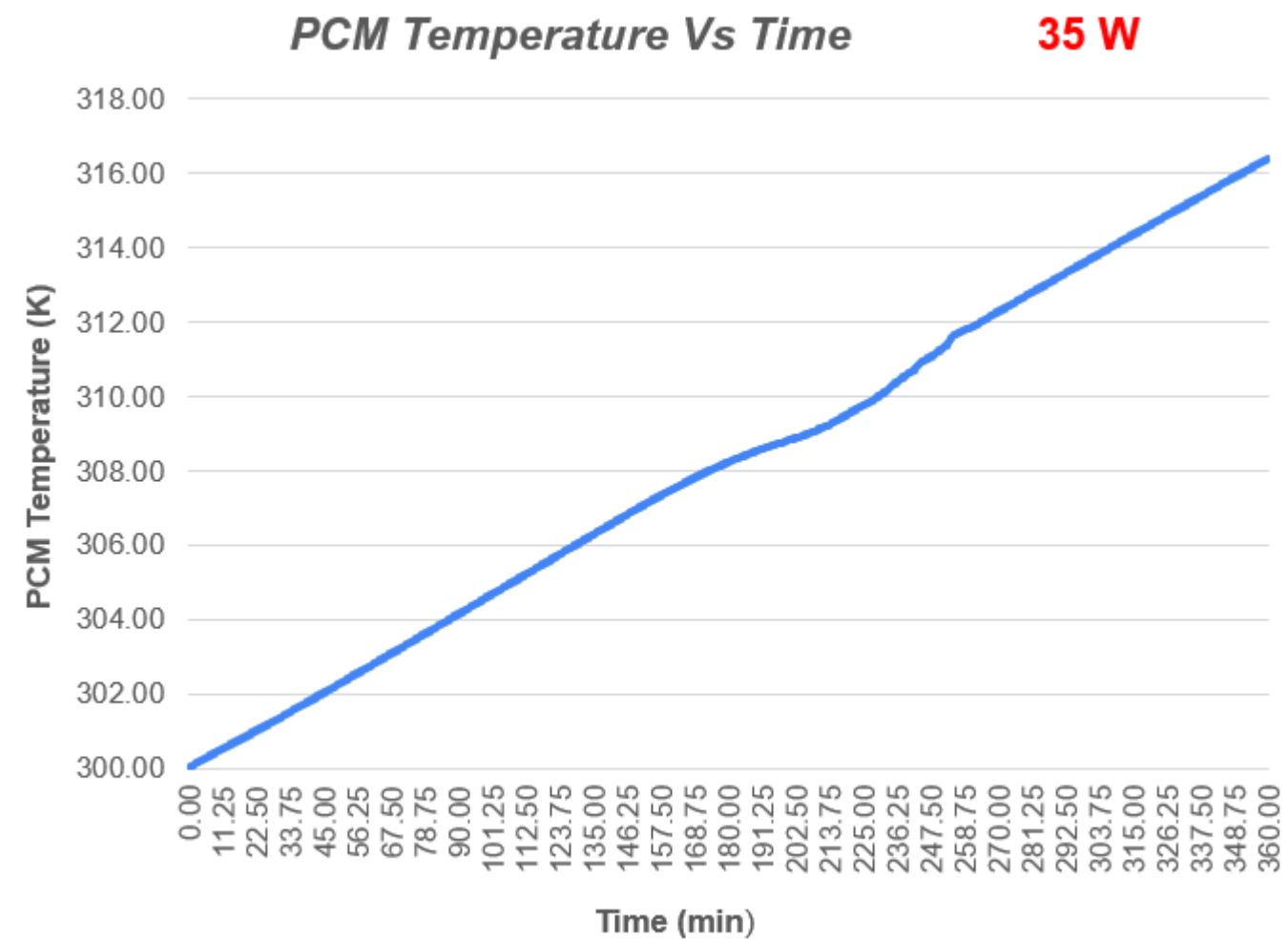


Fig. PCM Temperature vs Time graph

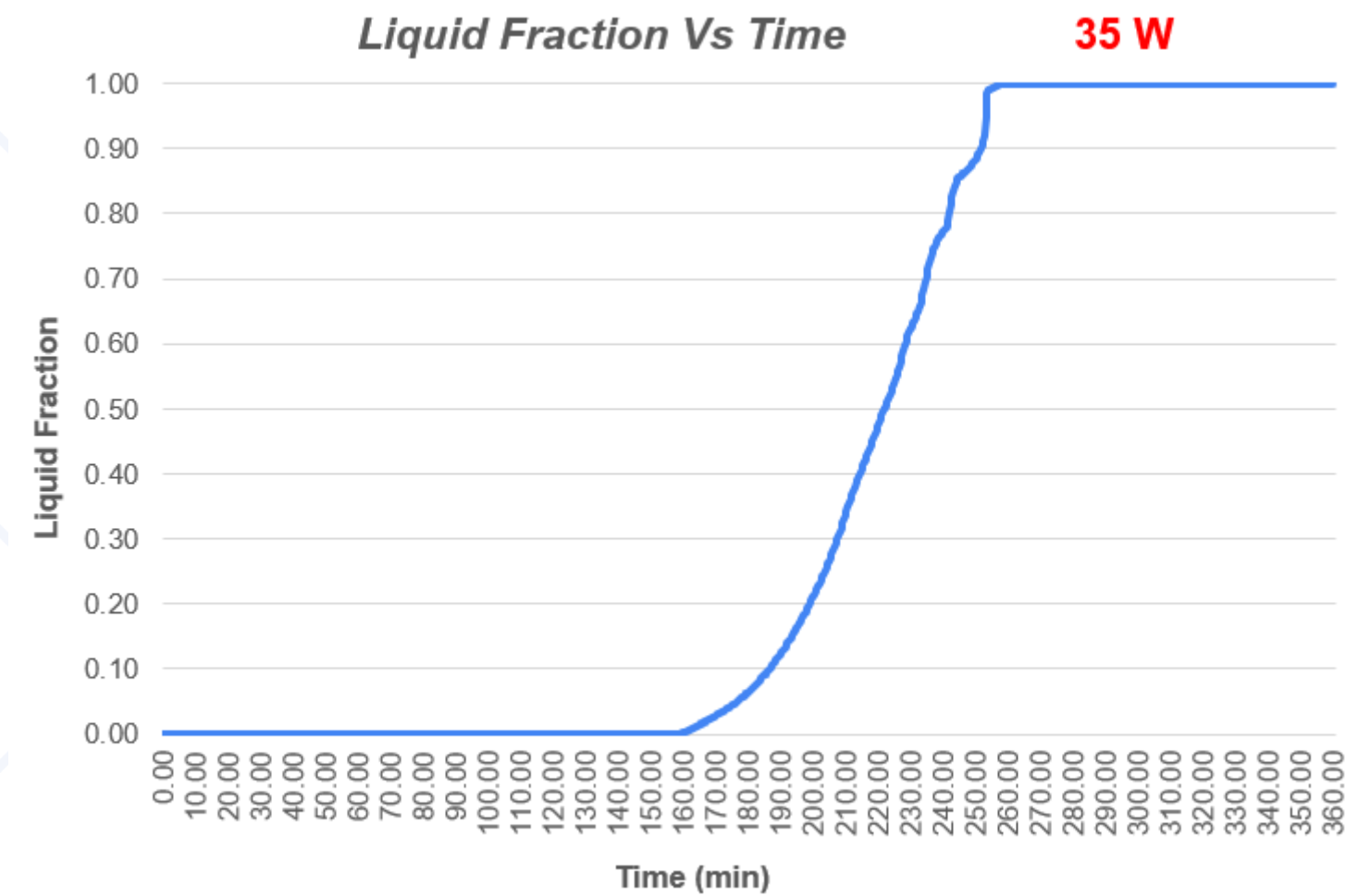


Fig. Liquid Fraction vs Time graph



Liquid Fraction Contours:

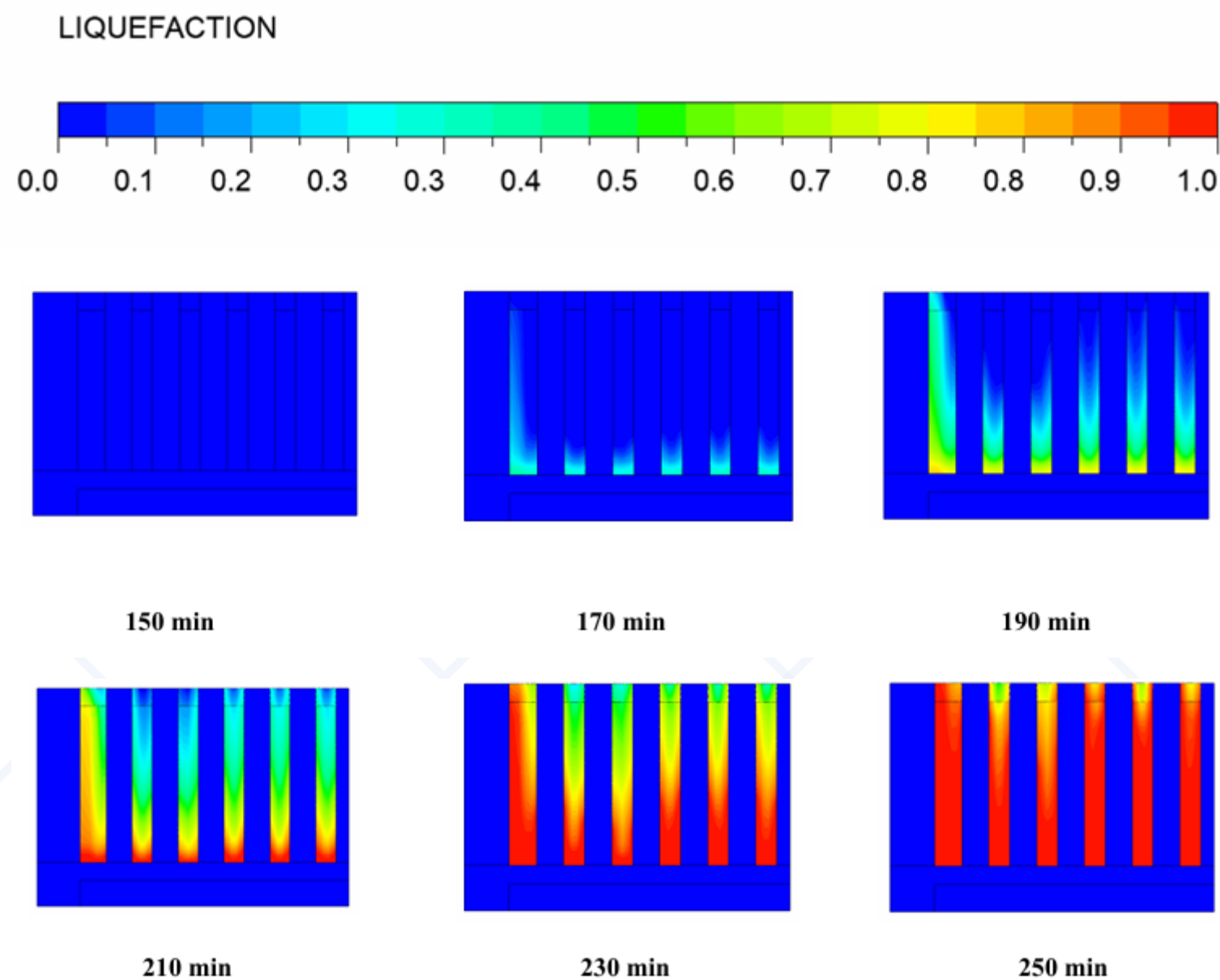


Fig. Liquid Fraction Contours (2D)

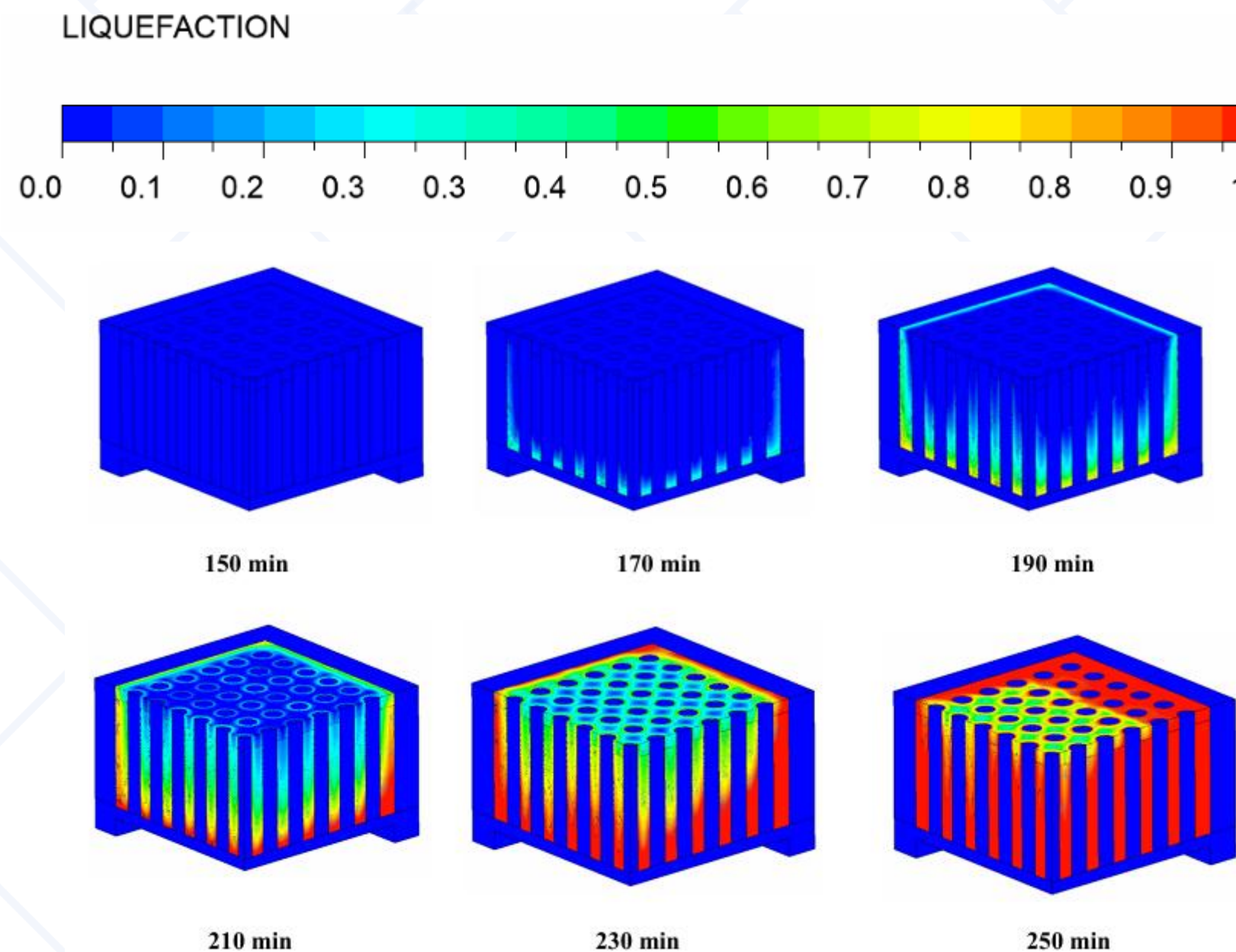


Fig. Liquid Fraction Contours (3D)



Contd.

Temperature Contours:

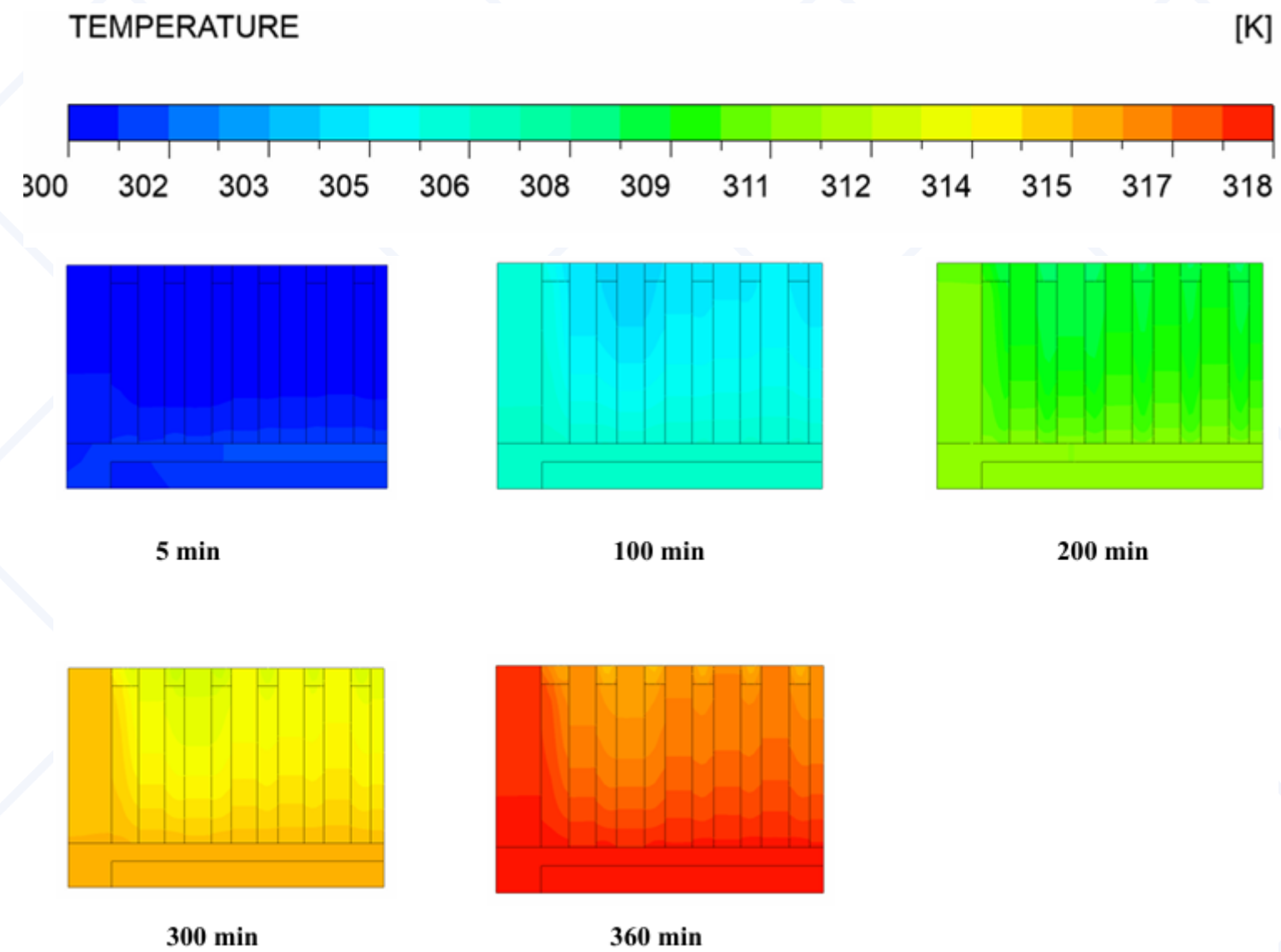


Fig. Temperature Contours (2D)

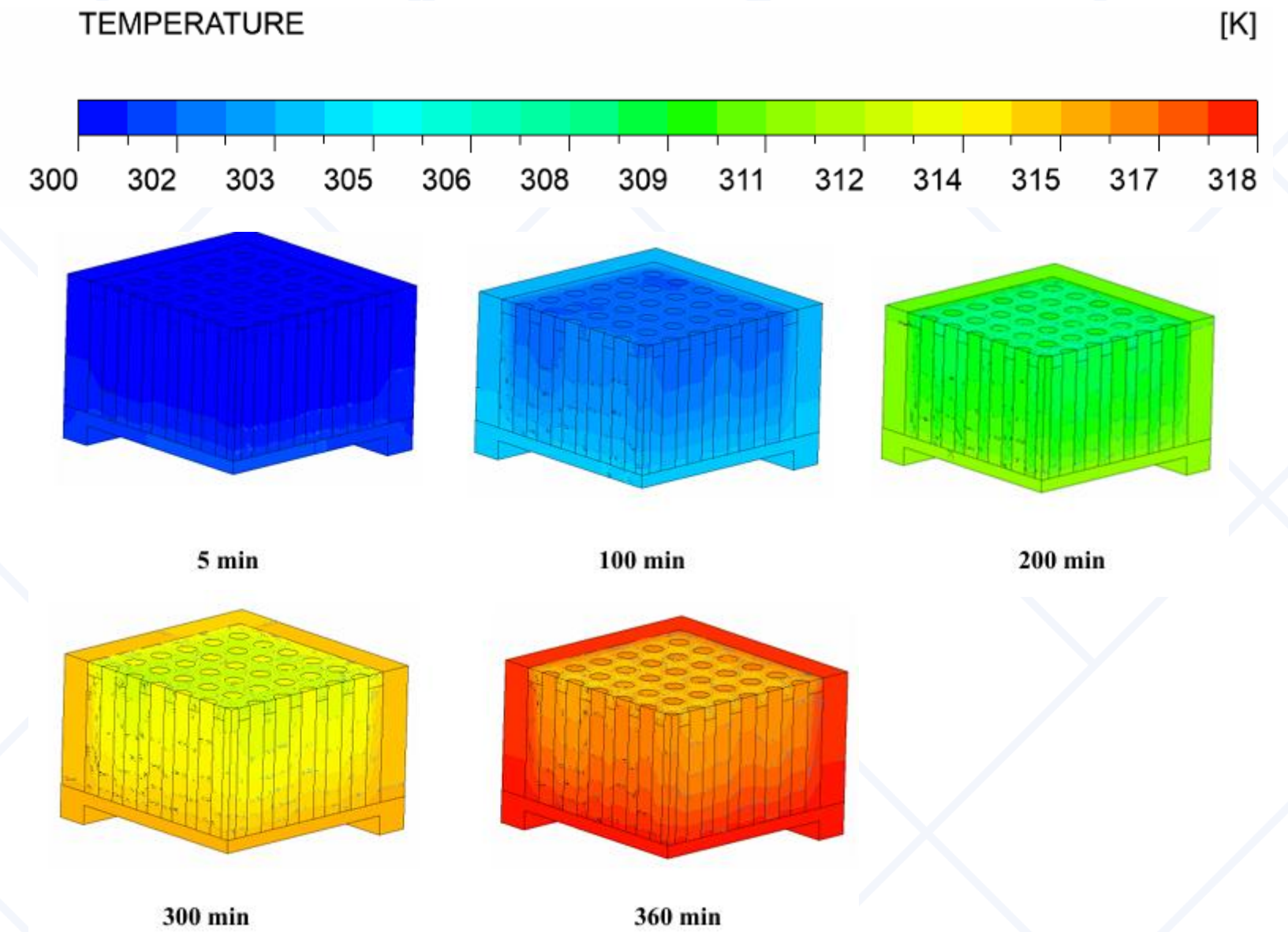


Fig. Temperature Contours (3D)



(4) COMPARISON BETWEEN THE CYLINDRICAL PIN FIN CONFIGURATION AT 25W, 30W & 35W:

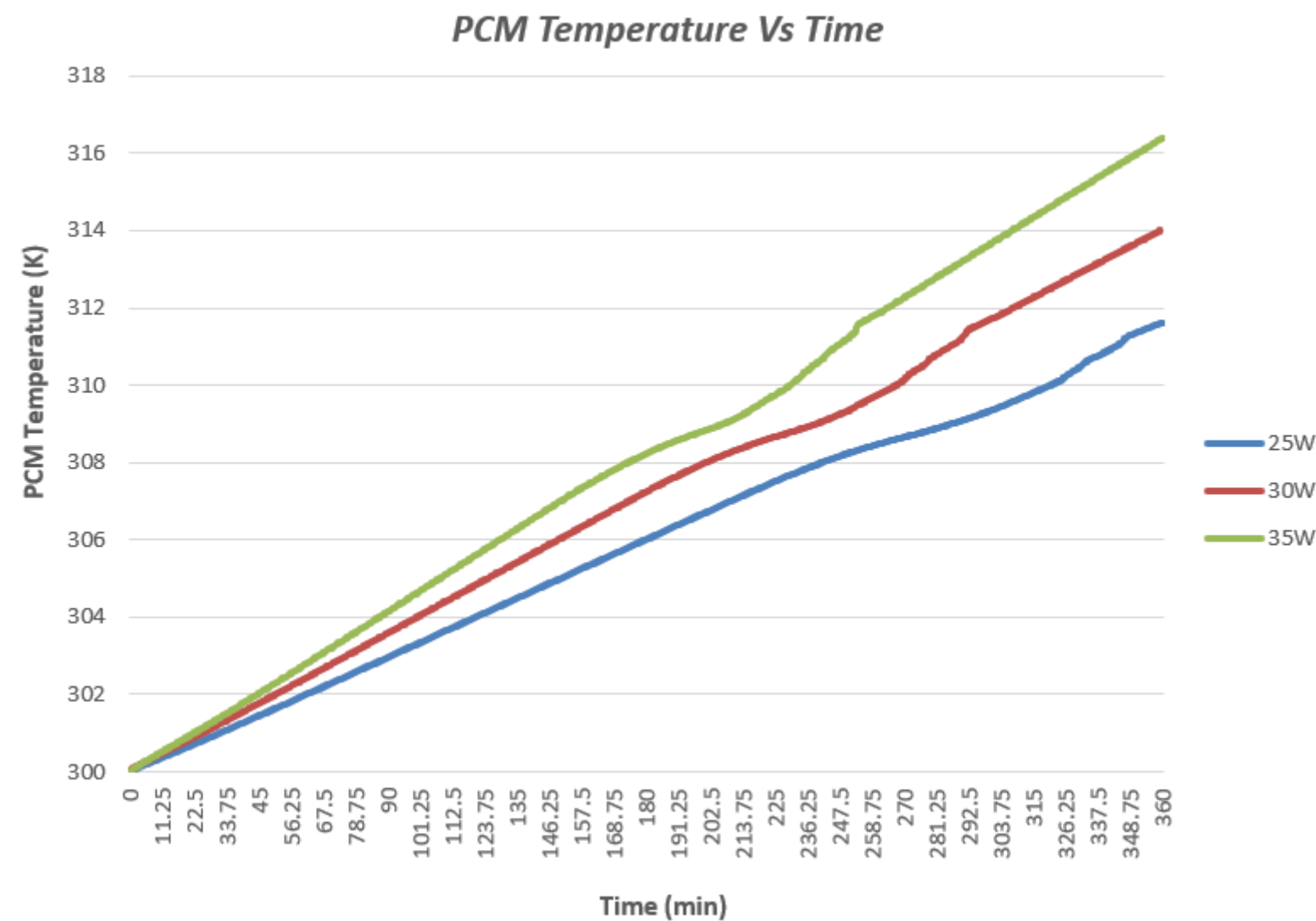


Fig. PCM Temperature vs Time graph

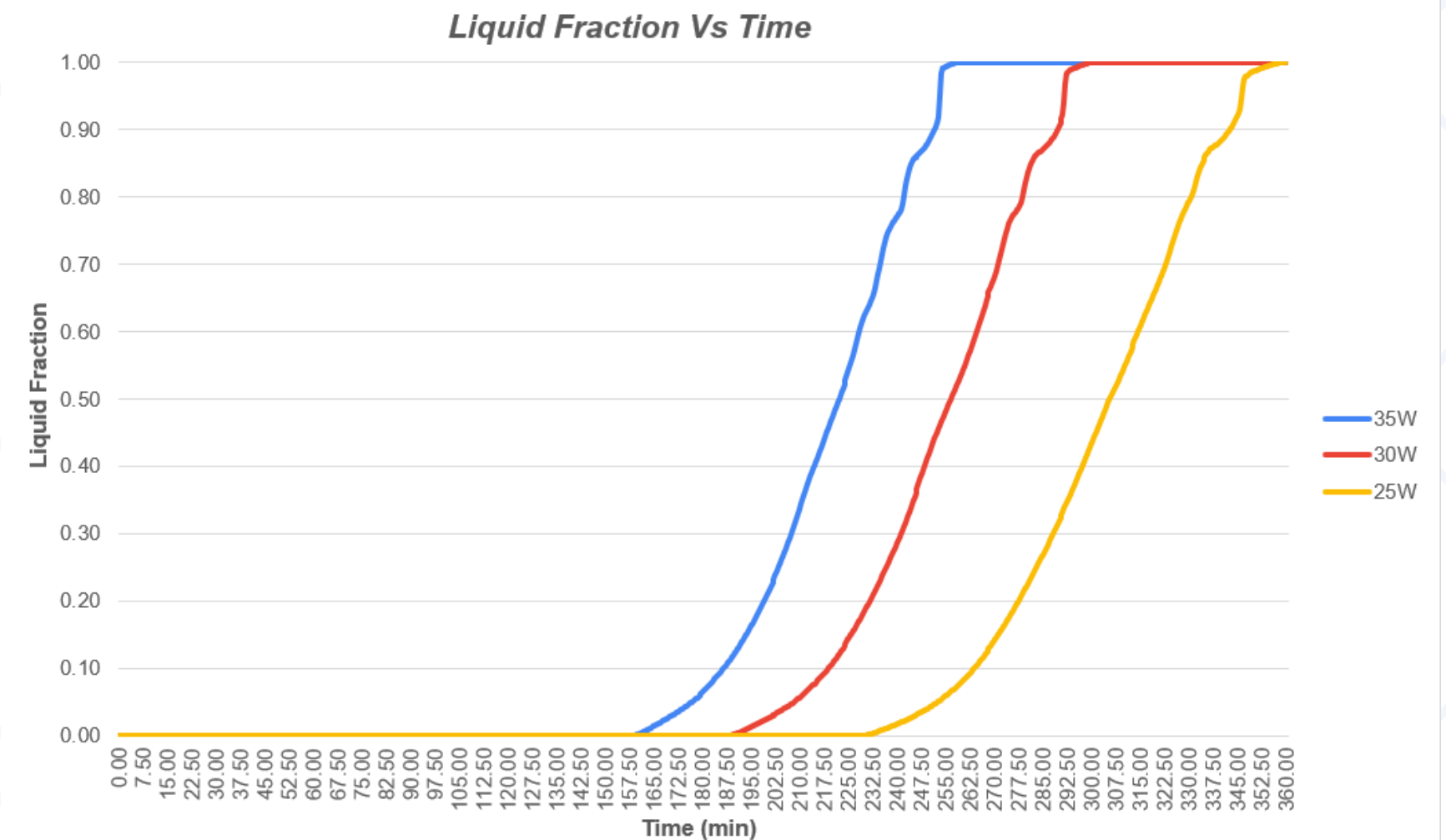


Fig. Liquid Fraction vs Time graph



Liquid Fraction Contours:

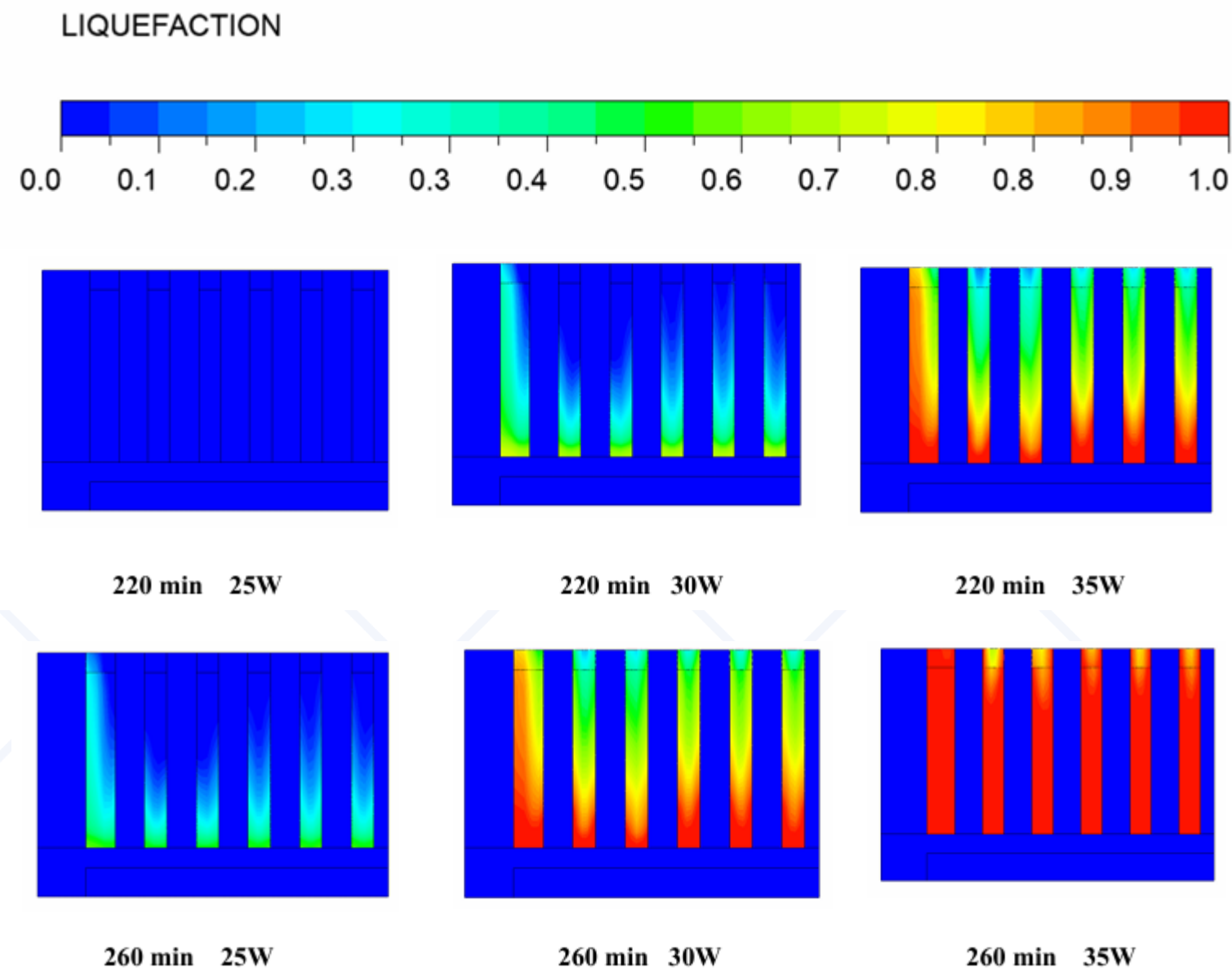


Fig. Liquid Fraction Contours (2D)

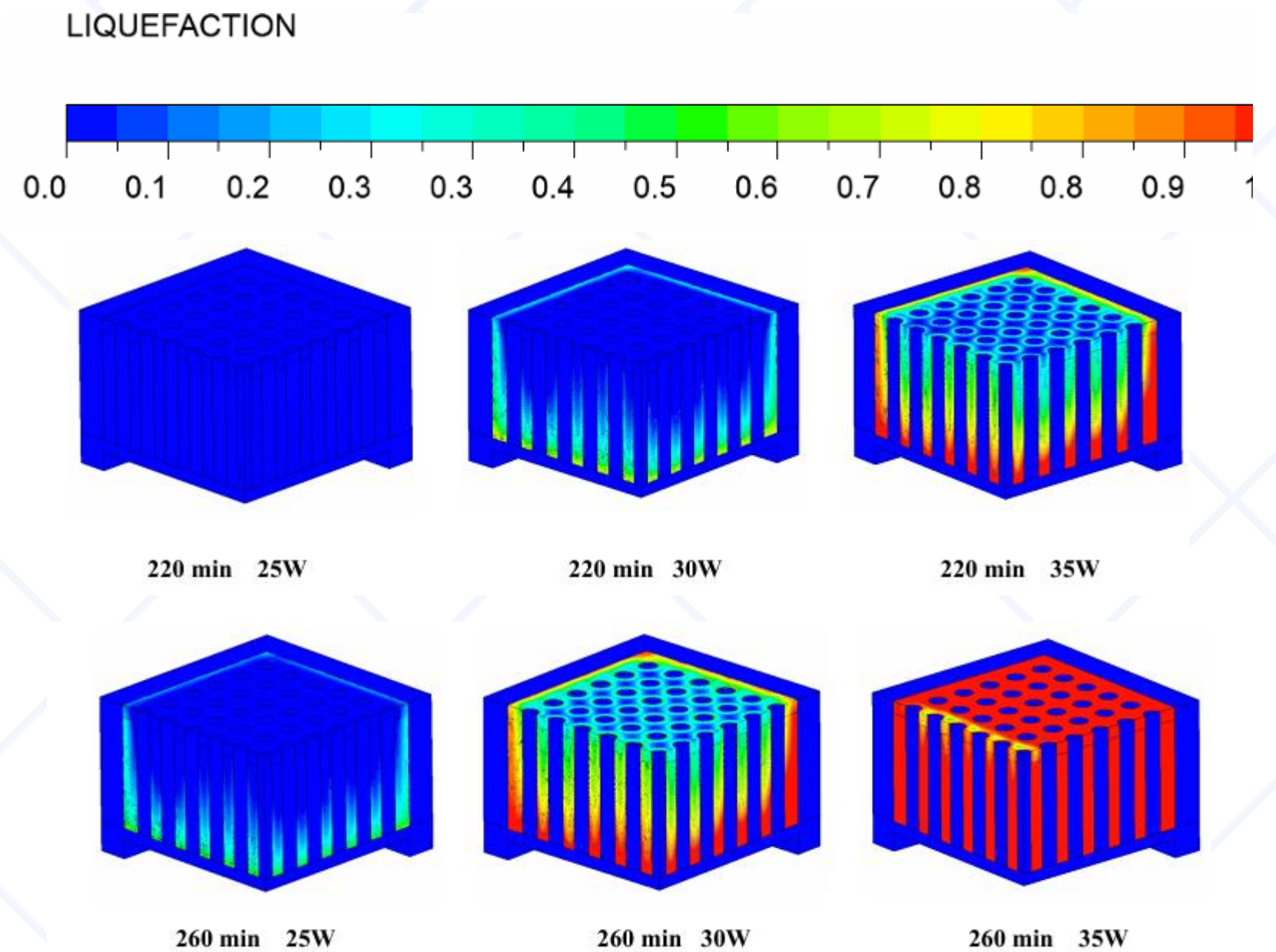


Fig. Liquid Fraction Contours (3D)



Liquid Fraction Contours:

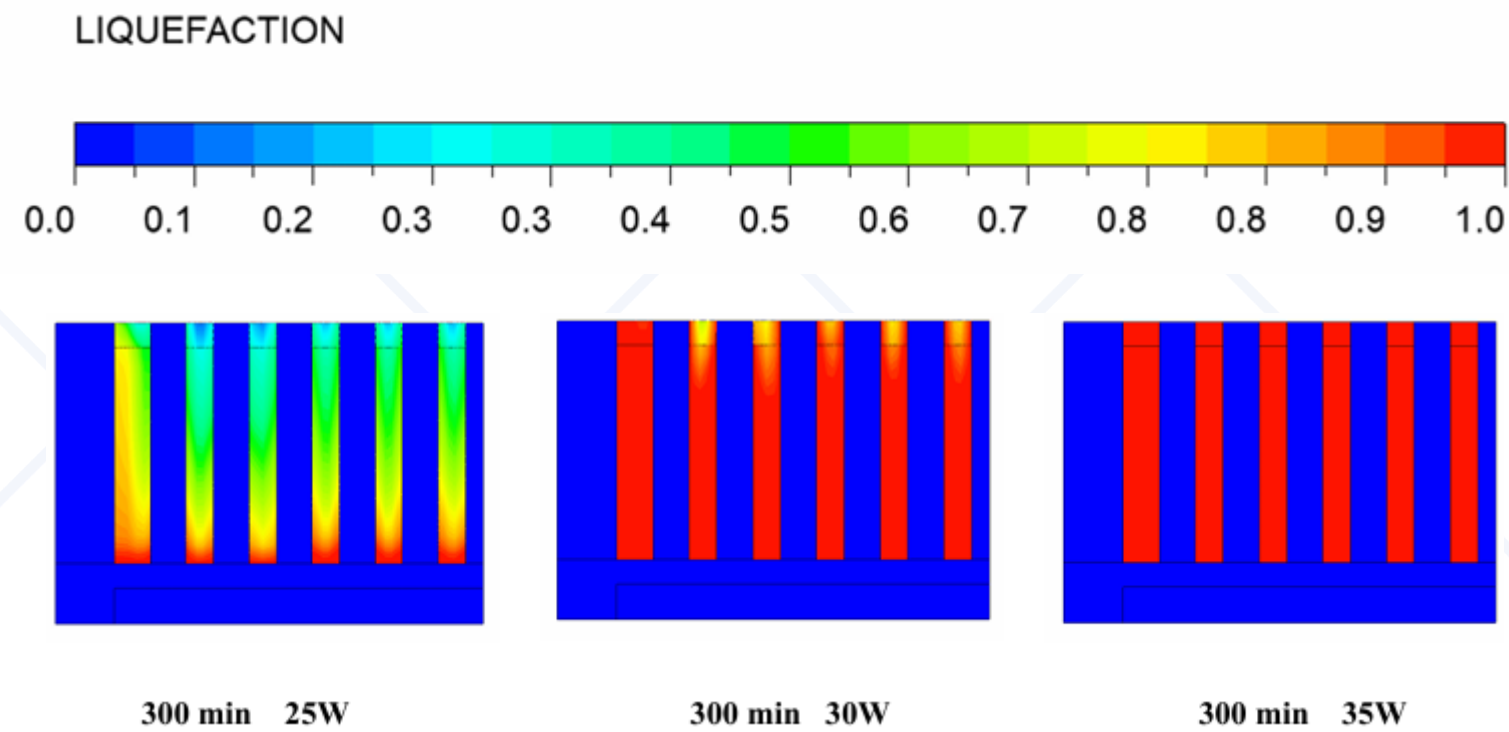


Fig. Liquid Fraction Contours (2D)

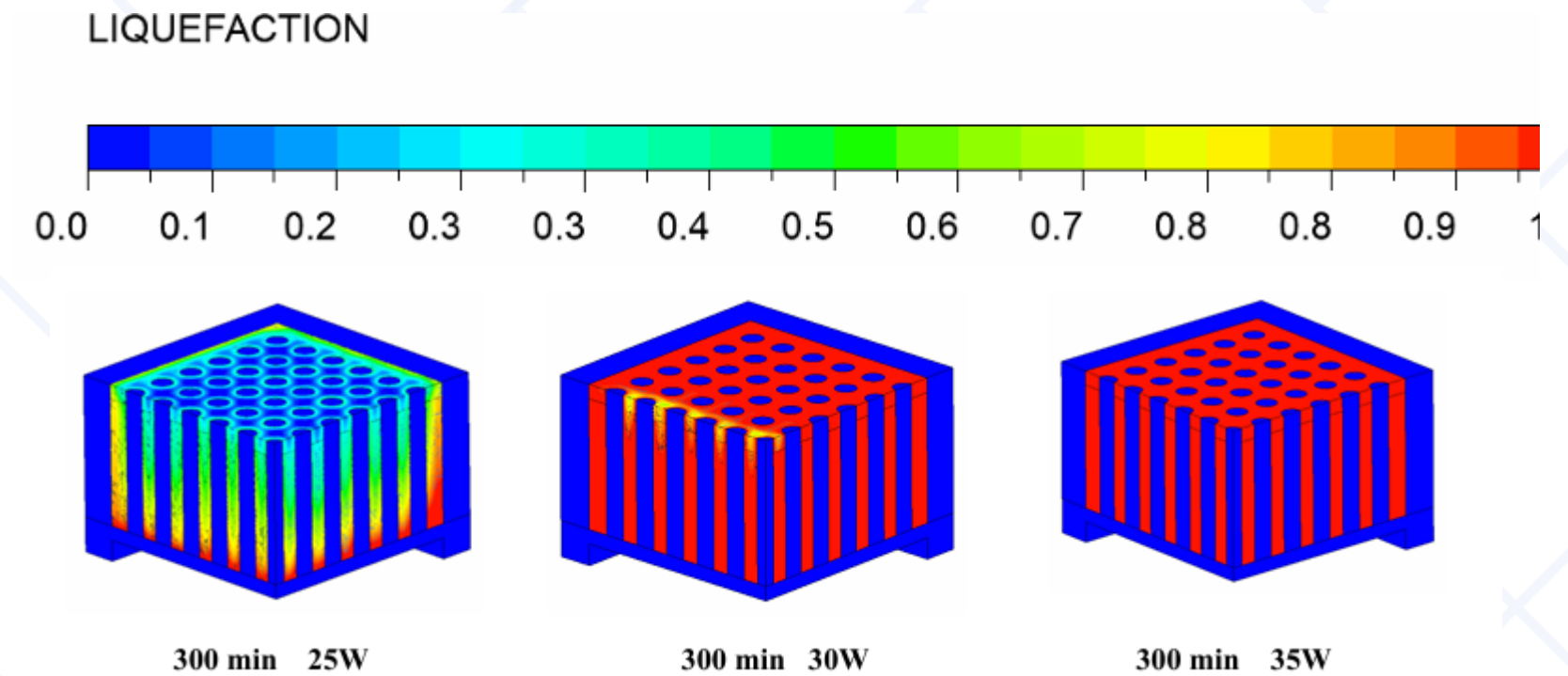


Fig. Liquid Fraction Contours (3D)



Temperature Contours:

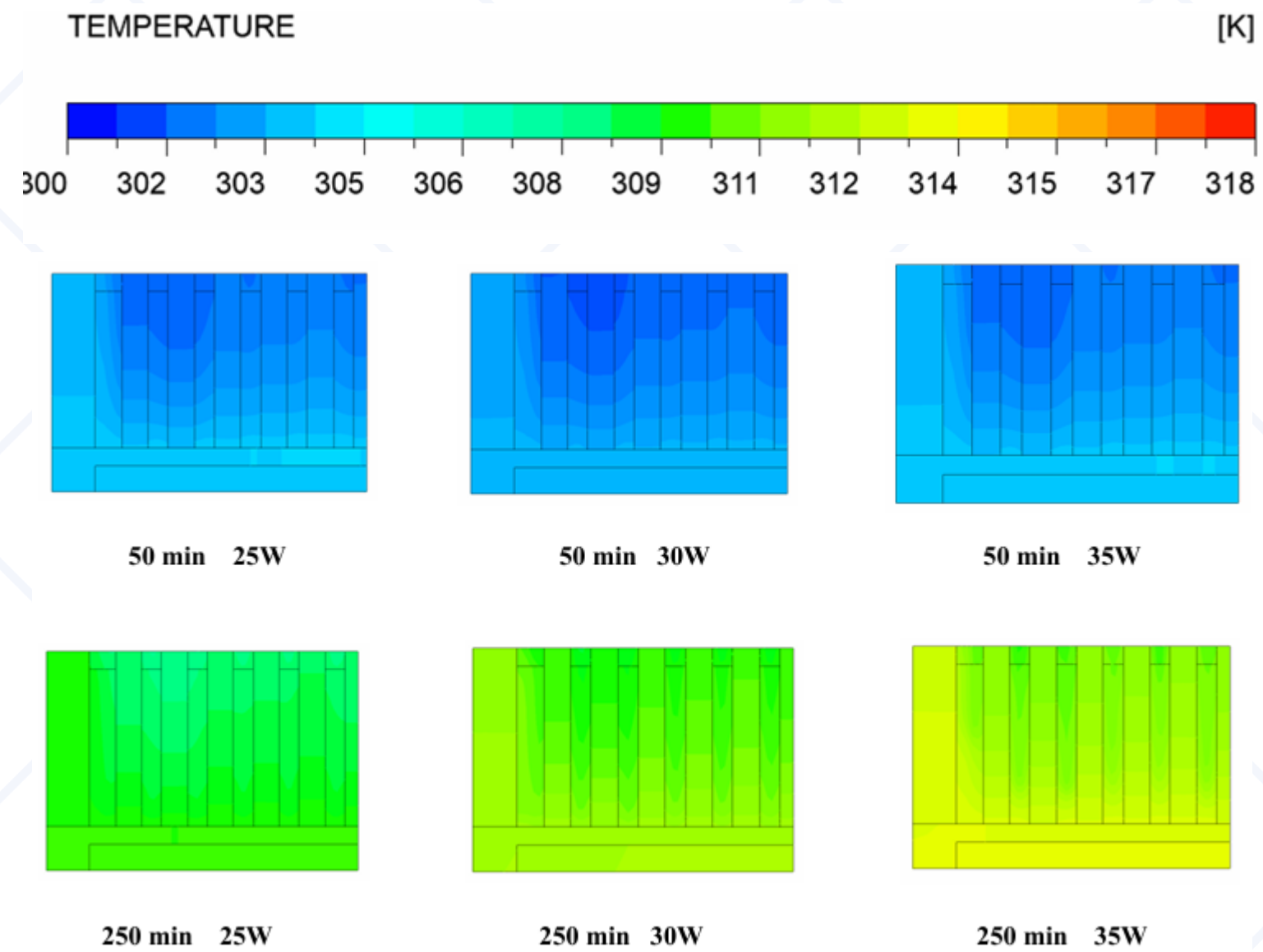


Fig. Temperature Contours (2D)

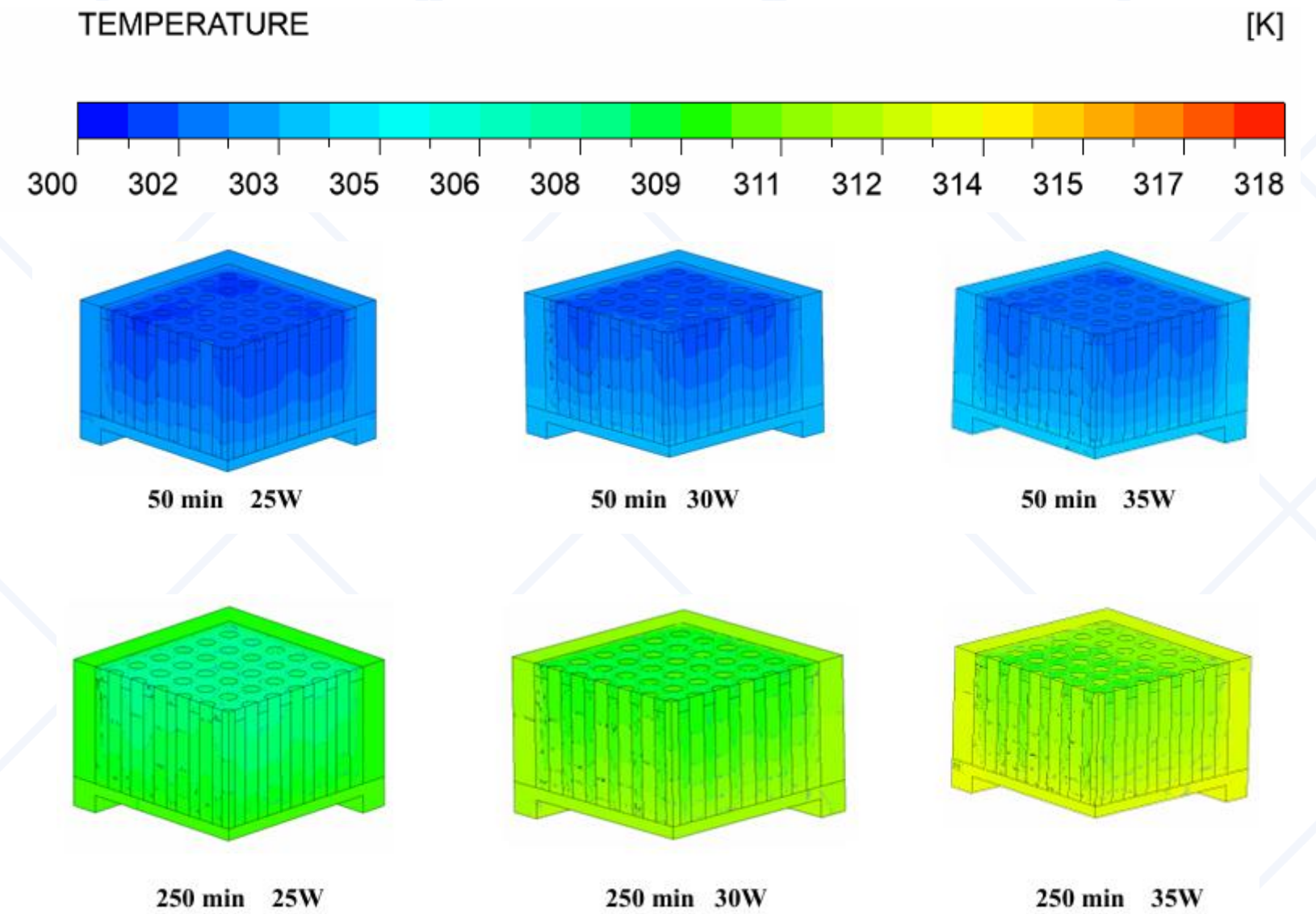


Fig. Temperature Contours (3D)



Contd.

Temperature Contours:

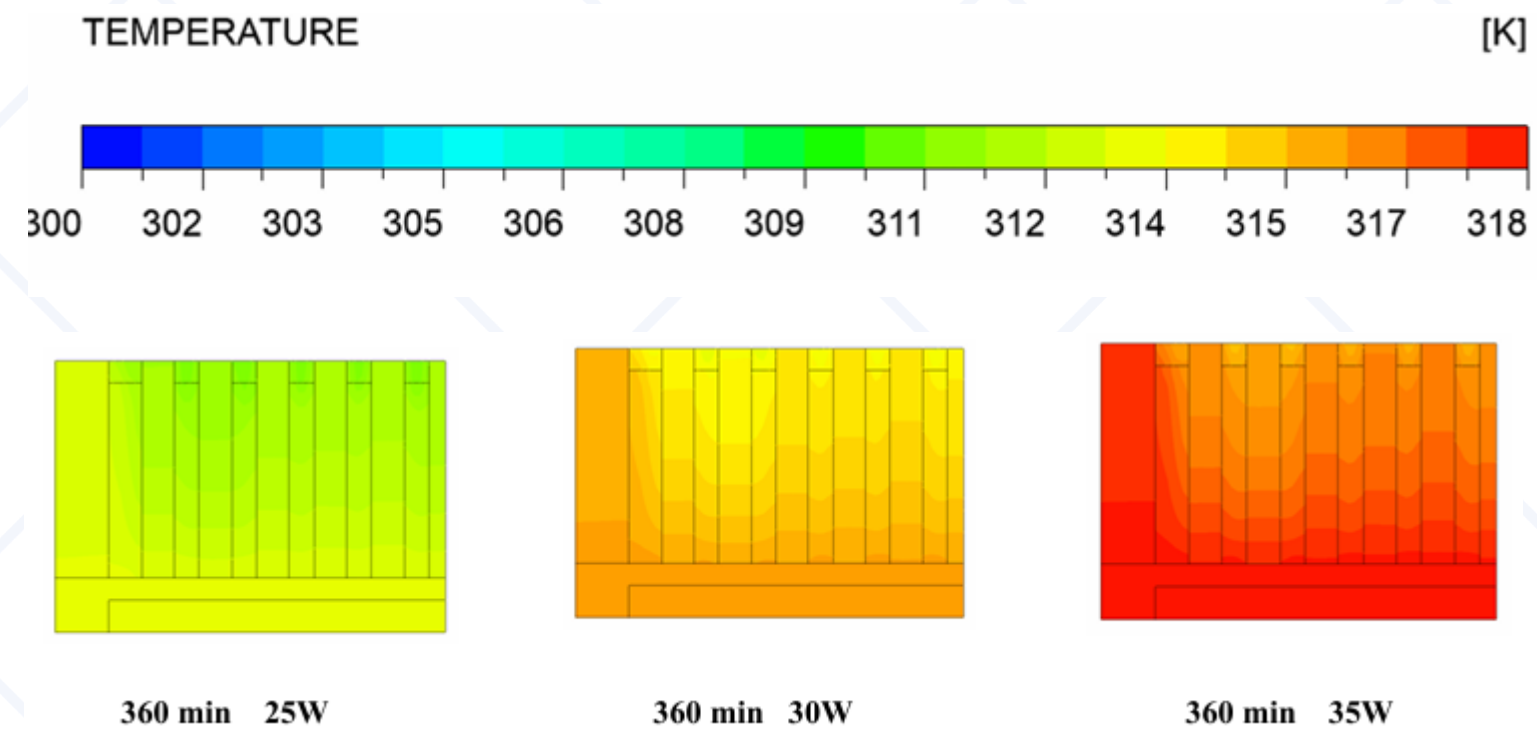


Fig. Temperature Contours (2D)

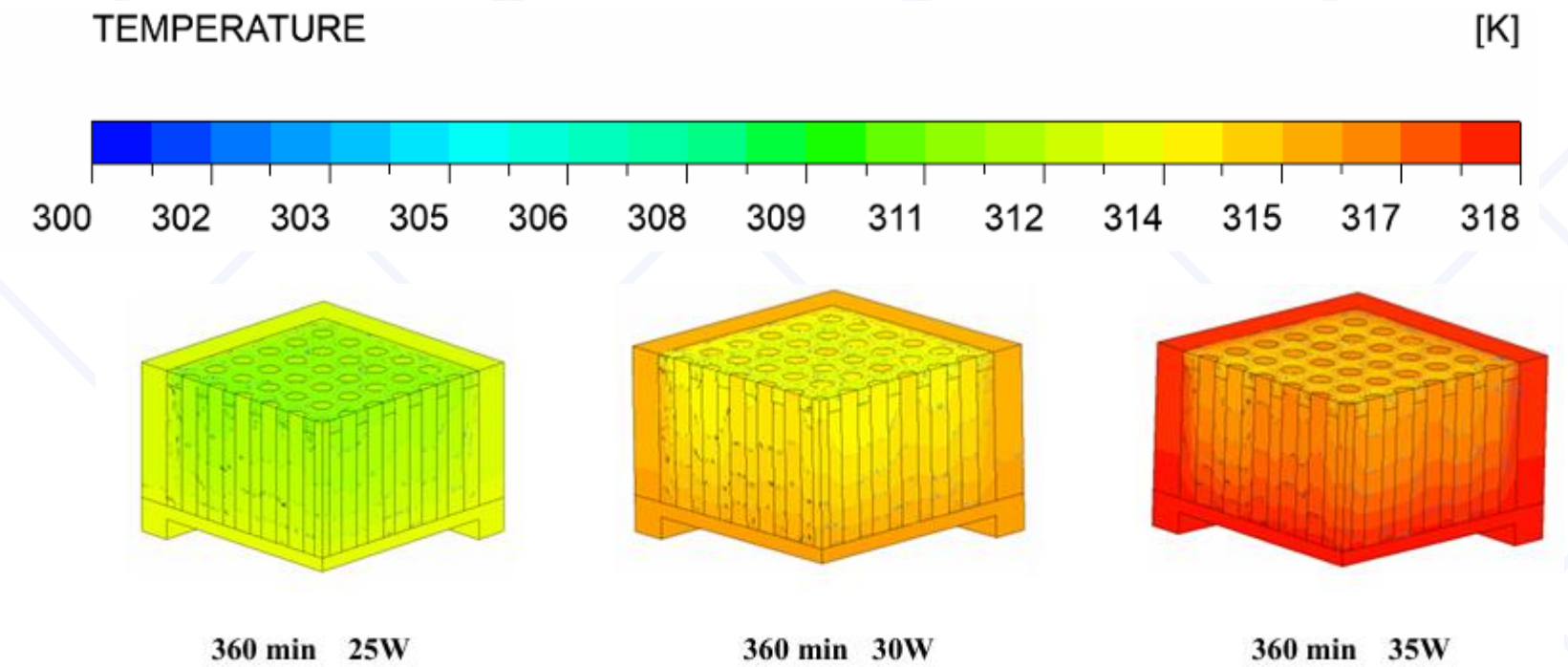


Fig. Temperature Contours (3D)



CFD MODEL AND VALIDATIONS:

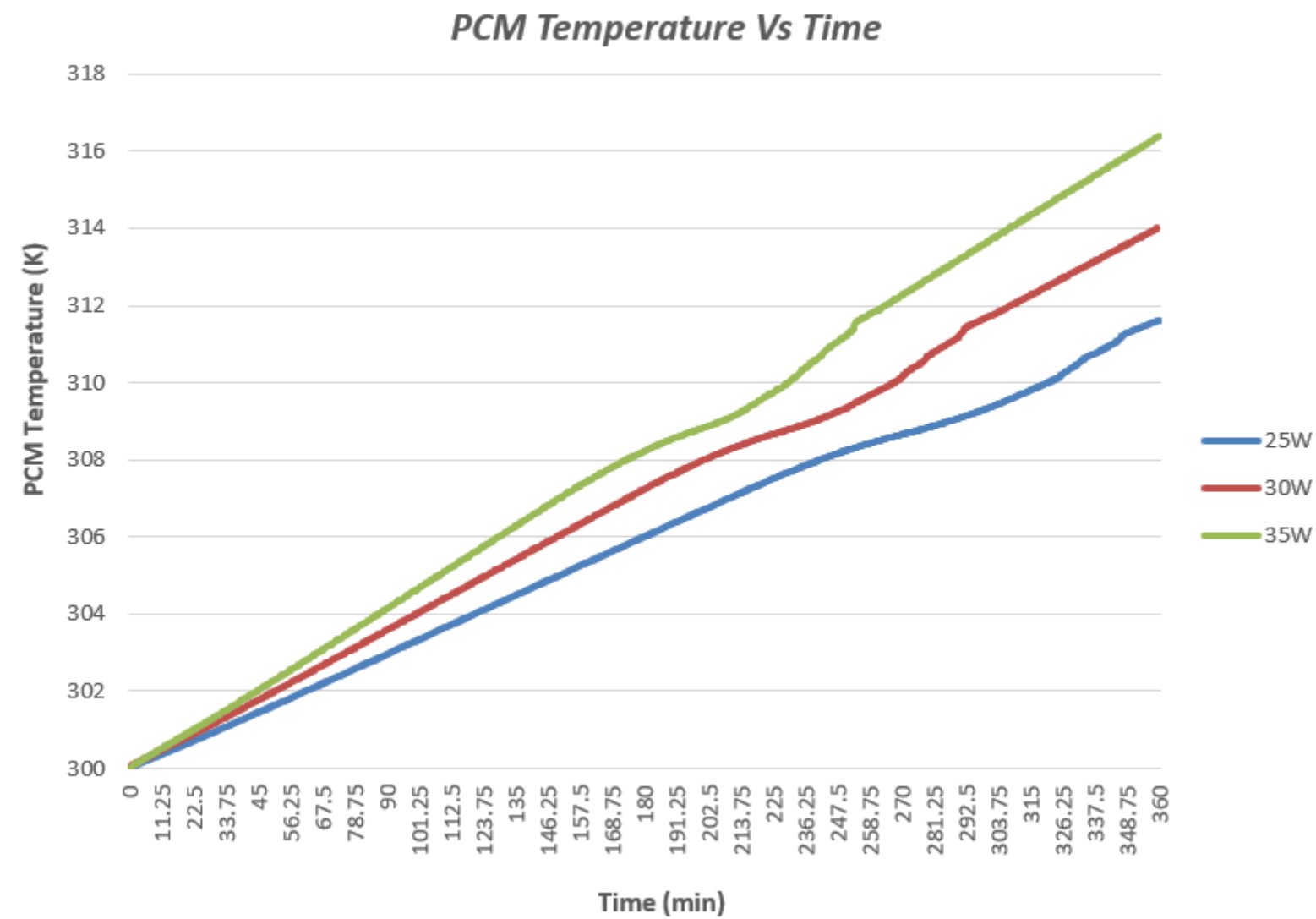


Fig. Temperature vs Time graph

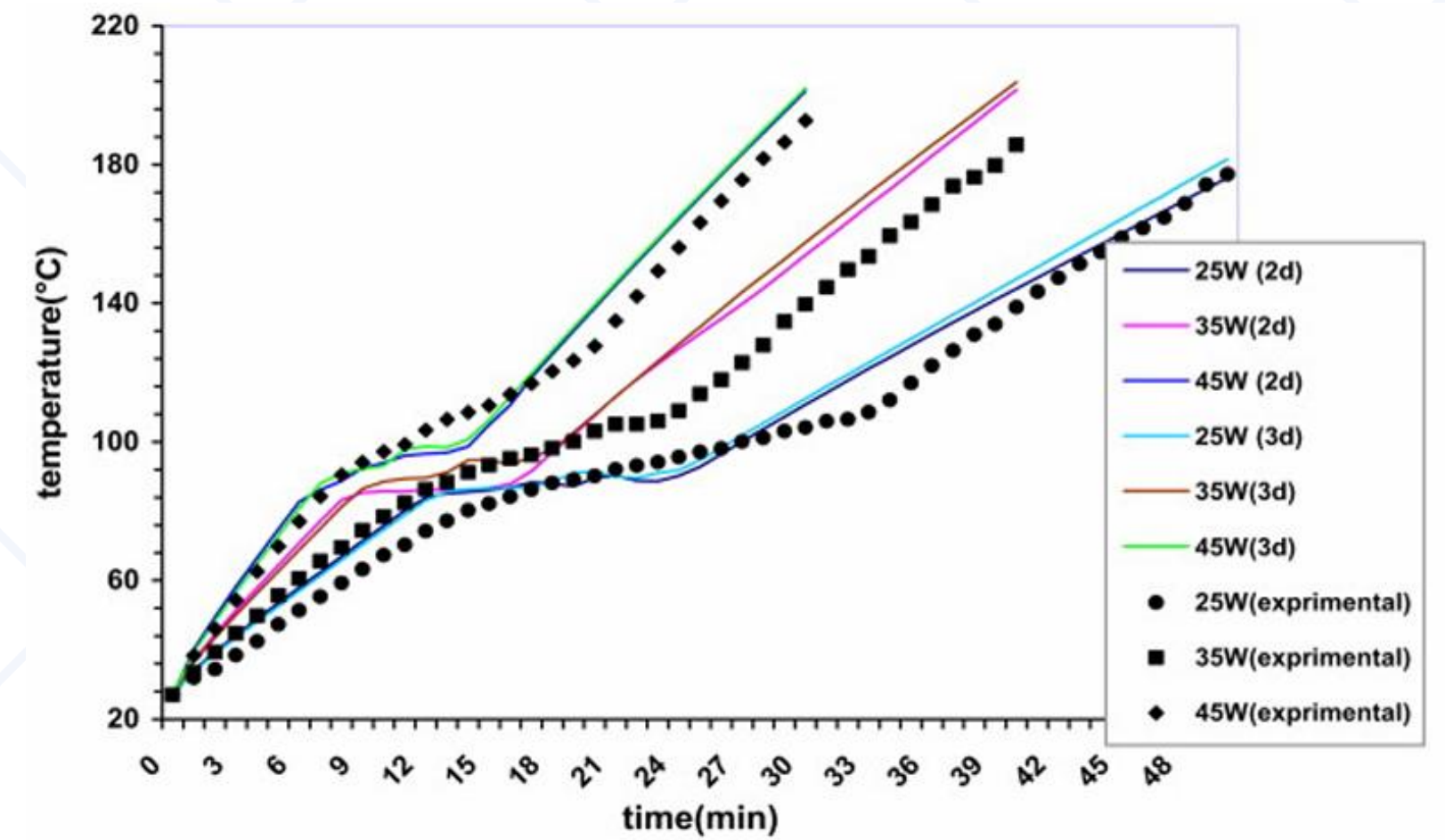


Fig. Temperature vs Time graph

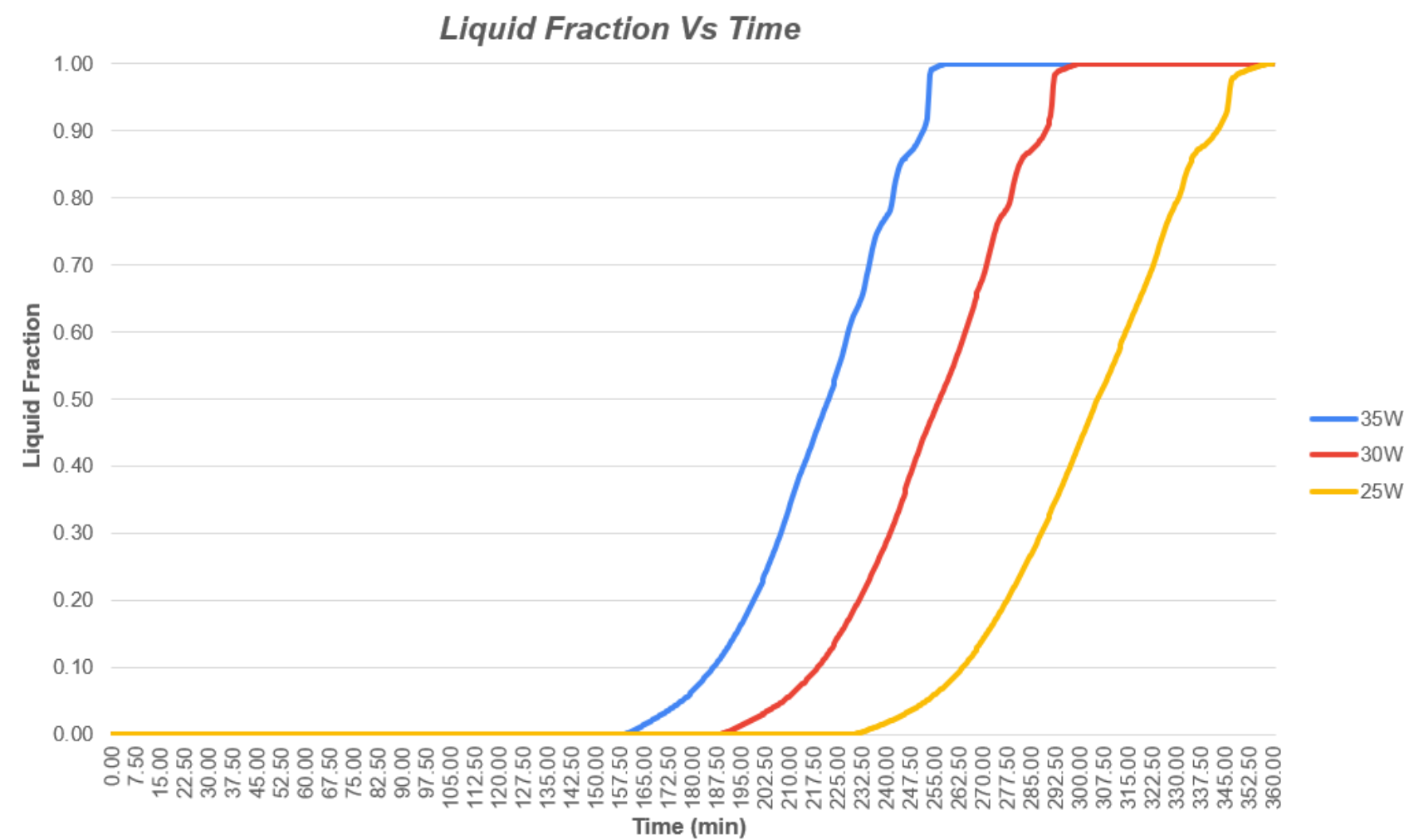


Fig. Liquid Fraction vs Time graph

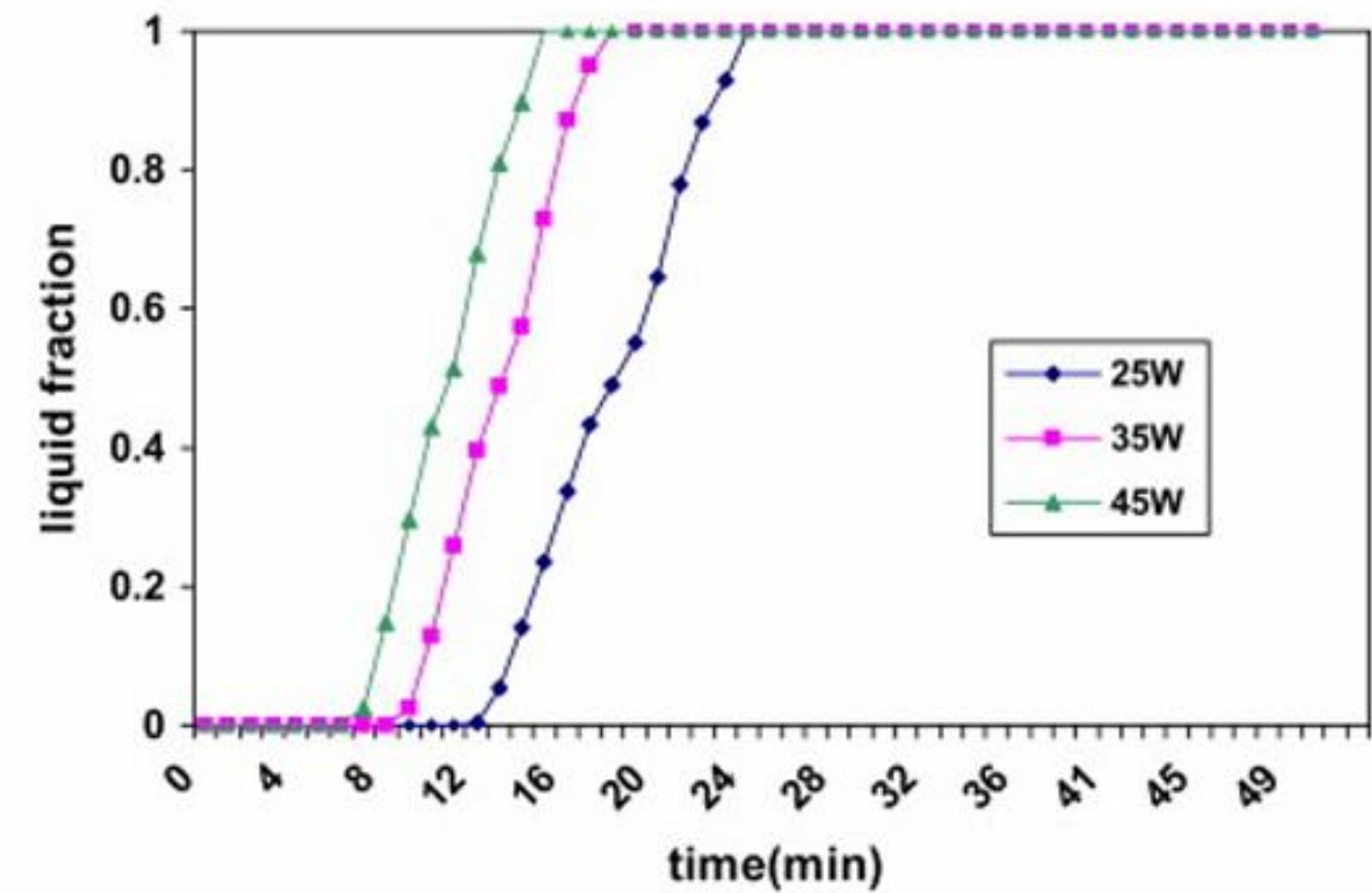


Fig. Liquid Fraction vs Time graph

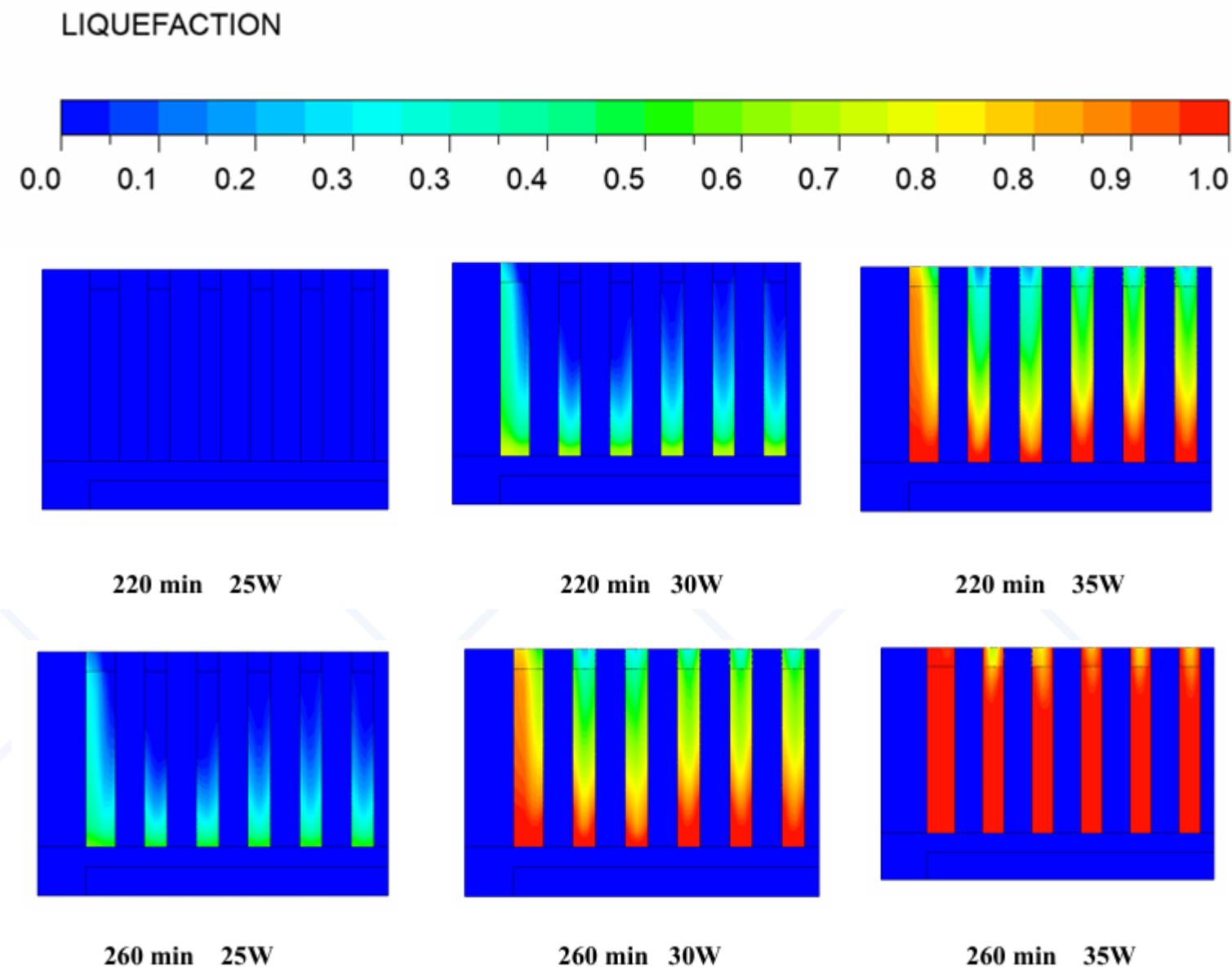


Fig. Liquid Fraction Contours

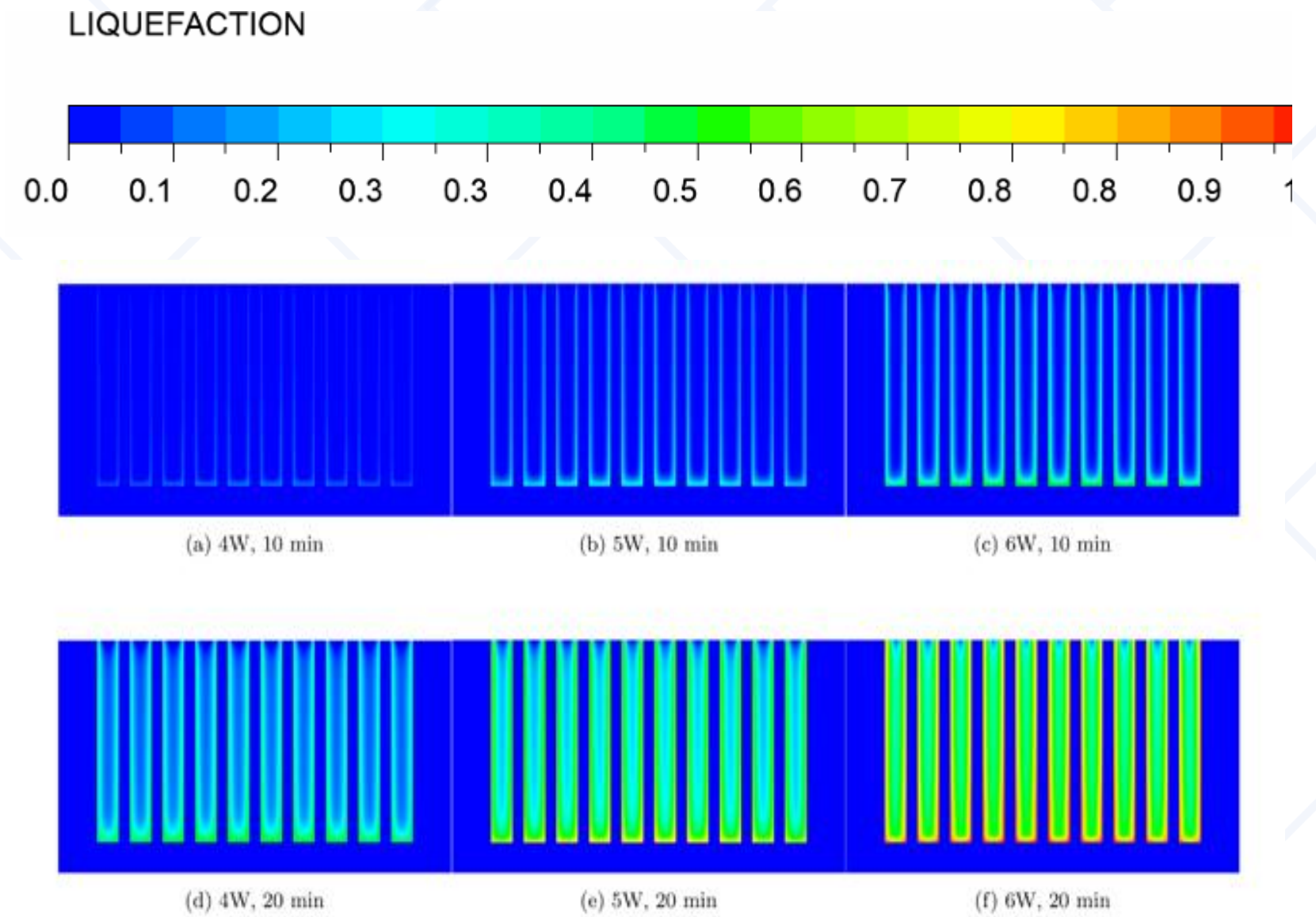


Fig. Liquid Fraction Contours



Conclusions

1. Phase Change Materials (PCMs) Enhance Thermal Performance,
2. Heat Input Directly Influences PCM Behavior
3. PCM Configuration Plays a Critical Role
4. Uniform Temperature Distribution Validated via Contour Plots
5. Robust Performance Across Boundary Conditions



Reference

01 <https://www.sciencedirect.com/science/article/pii/S0017931017349852>

02 <https://www.sciencedirect.com/science/article/pii/S0017931017328107>

03 <https://www.sciencedirect.com/science/article/pii/S2214157X21001763>

04 <https://www.sciencedirect.com/science/article/pii/S0017931024011165>



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Thank You